

The Shape of Jets

Rikab Gambhir

[Ba, Dogra, **RG**, Tasissa, Thaler; [2302.12266](#)]

[**RG**, Thaler, Wu; WIP]

[**RG**, Larkoski, Thaler; [2410.05379](#)]

[**RG**, Osathapan, Thaler; [2403.08854](#)]

[**RG**, Mastandrea; WIP]

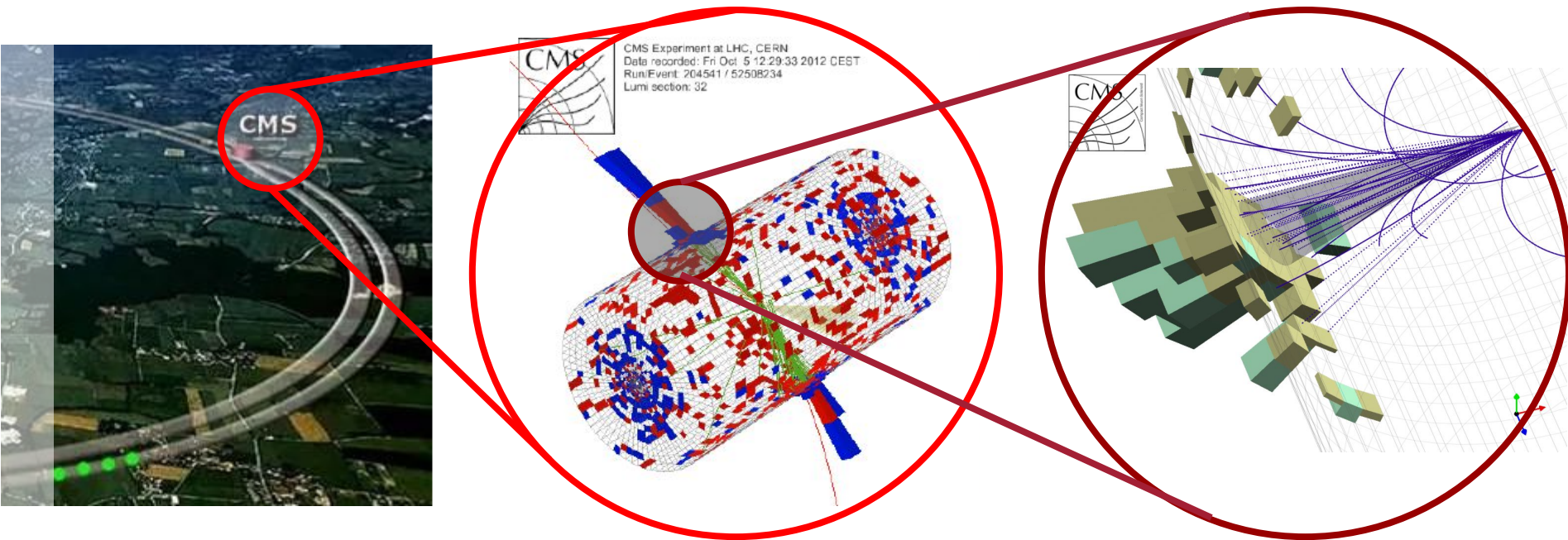
[Du Plessis, **RG**; WIP]

The building blocks of Life are
made up of shapes and spaces.

Joey Lawsin

“ quote fancy

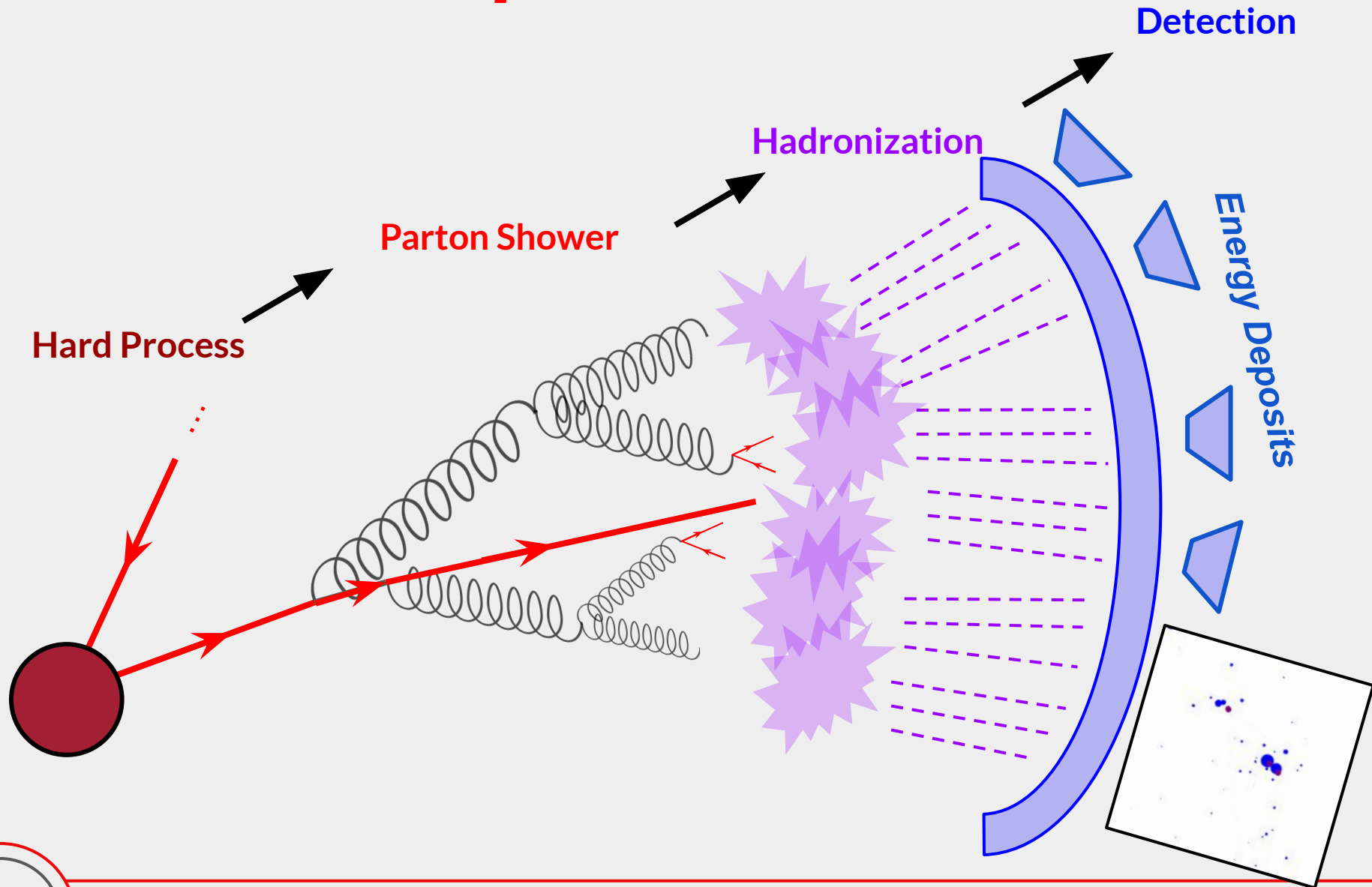
Our Setting: **Jets** at the **LHC**



My end goal: understand these better!

*Not literally the same jet

Jets are complicated



Jets are complicated

Detection

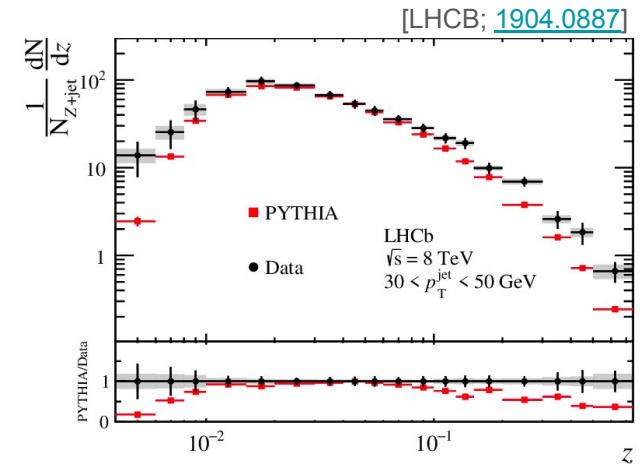
Hadronization

Jets are messy collections of 100's of particles with subtle latent correlations.

Hard to make predictions for the full distribution — *even before worrying about detector effects!**

Can we come up with some simple “summary statistics” (**Observables**) that characterize jets and are easier to work with?

Today's Talk: **Yes**.

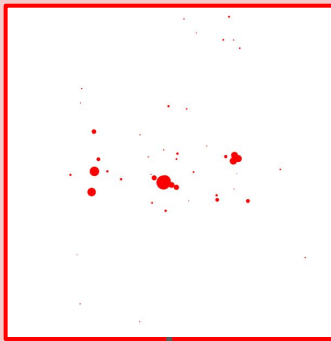


Example: PYTHIA struggles to reproduce charged hadron momentum fractions in jets at LHCb

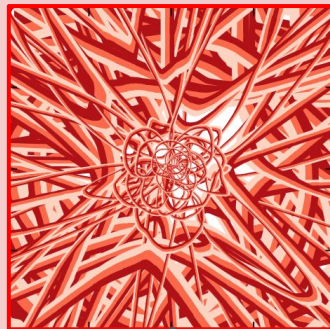
*I won't discuss detector effects in this talk much. But please ask me more about using ML to calibrate later!

Things about jets a **machine** can understand

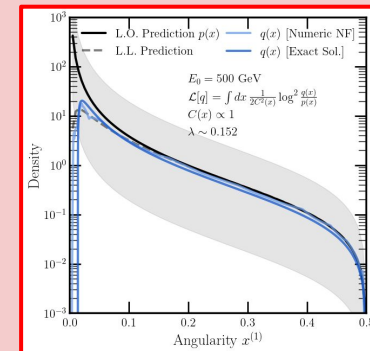
Complicated point clouds



Huge Latent Spaces

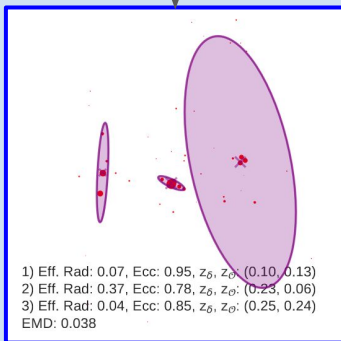


Nontrivially constrained function spaces

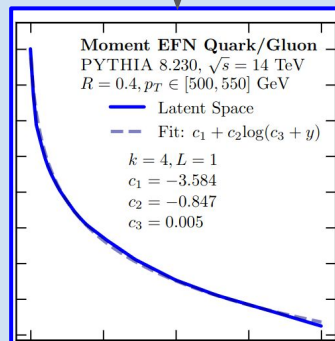


Interface: Targeted and goal-motivated ML function spaces and loss functionals!

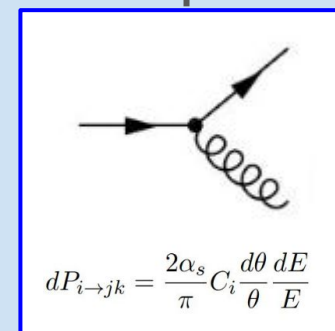
Using faithful
optimal transport



Using Moment
Pooling



Using Re-Normalizing
Flows



Basic Kindergarten Shapes

Addition, multiplication, and a
few elementary functions

Tree-level Feynman
Diagrams

Things about jets my **mortal physicist brain** can understand

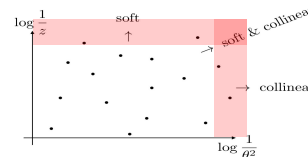
Outline

This talk is going to be **several mini-vignettes** about applying geometry and ML to jet physics.

Key takeaway: By enforcing physical principles (IRC-safety, unitarity, etc) and keeping your end goal in mind (robustness, simplicity, interpretability, etc), we can design loss functionals and models to help us encode precisely the physics we want to learn! We have *full control* over our physics.

Part 0 (You are Here)

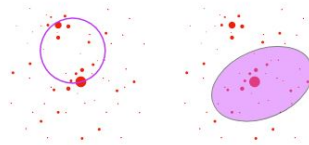
Intro: IRC-Safety, Jets, and Geometry



[Ba, Dogra, **RG**, Tasissa, Thaler, [2302.12266](#)]

Part I

Can you Hear the Shape of a Jet?



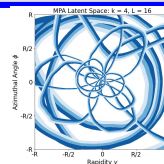
[Ba, Dogra, **RG**, Tasissa, Thaler, [2302.12266](#)]

[**RG**, Thaler, Wu, WIP]

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Part II

Moments of Clarity: Streamlining Latent Spaces



[**RG**, Osathapan, Thaler, [2403.08854](#)]

Part III

Renormalizing Flows: Taming Perturbation Theory

[**RG**, Mastandrea, WIP]

$$\exp\left[\alpha \frac{d}{d\alpha}\right] p(x) = p^0(x) + \alpha^1 q^1(x) + \dots + \alpha^N p^N(x) + \mathcal{O}(\alpha^{N+1})$$

equal!

$$\exp\left[\alpha \frac{d}{d\alpha}\right] q(x) = q^0(x) + \alpha^1 q^1(x) + \dots + \alpha^N q^N(x) + \mathcal{O}(\alpha^{N+1})$$

Part 0

Intro: **IRC-Safety, Jets, and Geometry**

~~THE UNREASONABLE EFFECTIVENESS
OF MATHEMATICS IN THE NATURAL
SCIENCES~~

Eugene Wigner

The Wasserstein Metric

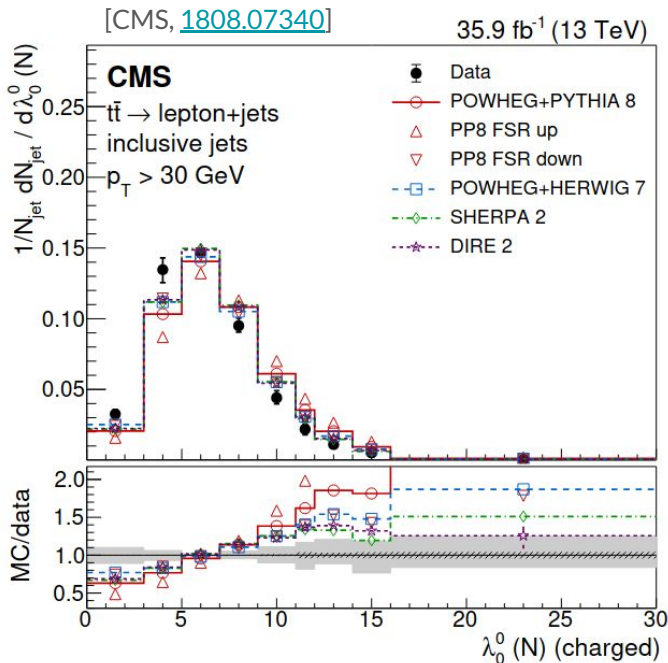
Collider Physics

Mathematics, right or wrong, has no appeal to any part of our weaker nature, without the gorgeous trappings of painting or music, yet sublimely pure, and capable of a stern perfection such as only the greatest art can show. The true spirit of delight, the exaltation, the sense of being more than Man, which is the touchstone of the highest excellence, is to be found in mathematics as surely as in poetry.

- BERTRAND RUSSELL, Study of Mathematics

Primer: The **IRC** Divergence

We want to predict* the collider output (or at least, some of it).



Lets try calculating the number of particles in a collision!

$$\frac{d\sigma}{dN} = \sum_{N'=0}^{\infty} \sum_{\text{QM}\#\text{'s}} \int d\Pi_{N'} |\mathcal{M}_{p_1 \dots p_{N'}, \text{QM}\#\text{'s}}|^2 \delta(N - N')$$

$$\xrightarrow{N=3} \frac{2\alpha_s C_F}{2\pi} \log\left(\frac{1}{\Lambda_{IR}}\right) \log\left(\frac{1}{\theta_{\text{cut}}}\right) \quad \mathbf{\times}$$

Lesson: You can't predict everything – the **IRC** (infrared and collinear) divergence spoils it!**

*By “predict” in this talk, I mean in perturbative first-principles QCD, with no additional empirical models

**This IR-divergence structure is a generic feature of 4D Yang-Mills. Cannot be cured by renormalization

Why IRC-Safety?

An observable is **IRC-Safe** if soft or collinear splittings only change the observable in a continuous way.

- IRC-safety ensures observables are well-defined in perturbative QCD.
- IRC-safety ensures effects due to soft or collinear physics, such as hadronization, are suppressed and therefore easier to deal with.
- IRC-safety aids in experimental robustness – missing soft particles, finite resolution effects, etc are suppressed.
- We will see that IRC-safe observables relate to intuitive geometric properties of jets.
- We will see that huge classes of IRC-safe observables are easy to define and evaluate programmatically.

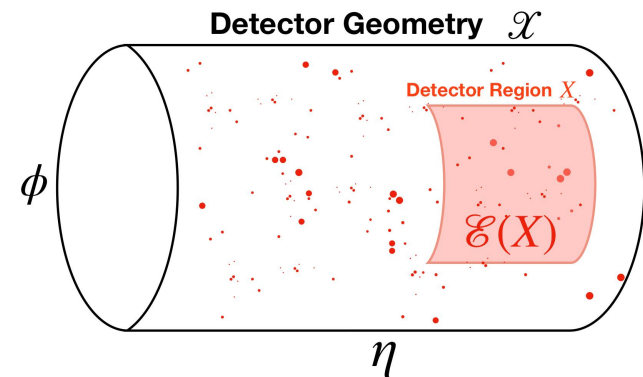
The Energy Flow

What is IRC-safe? Embed event data into the **energy flow**:

$$\mathcal{E}(y) = \sum_i z_i \delta(y - y_i)$$

Detector Coordinate (η, ϕ)

Energy Fraction E_i / E_{Tot}



The **energy flow** contains *ALL* IRC-safe information about an event!

- Claim 1: The energy flow is a continuous embedding of events.
- Claim 2: An observable $\mathcal{O}(\mathcal{E})$ is IRC-safe if and only if it is continuous with respect to the *weak* topology* on energy flows.
- Claim 3: The Wasserstein metric is the only *faithful* distance-preserving metric on energy flows.

*What happened to charge, flavor, mass, and other particle information? Ask me!

**In this talk, will treat “jets” and “events” as the same thing – the difference here is unimportant

The Energy Flow

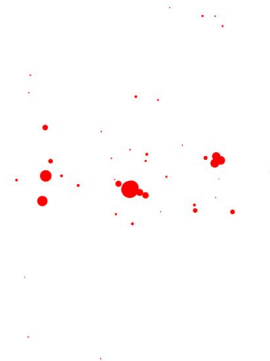
What is IRC-safe? Embed event data into the **energy flow**.

Emotionally, the IRC-safe part of a jet is just a **weighted point cloud**.

And weighted point clouds are just **probability distributions**.

So, asking IRC-safe questions about jets corresponds to asking questions about probability distributions!

What is the *mean* of this jet?
What is [*any moment*] of this jet?
How similar is this jet to [*given distribution*]?
How can I parametrize/fit this distribution?



A PYTHIA-simulated top jet as a point cloud

- Claim 1: The energy flow is a continuous embedding of events.
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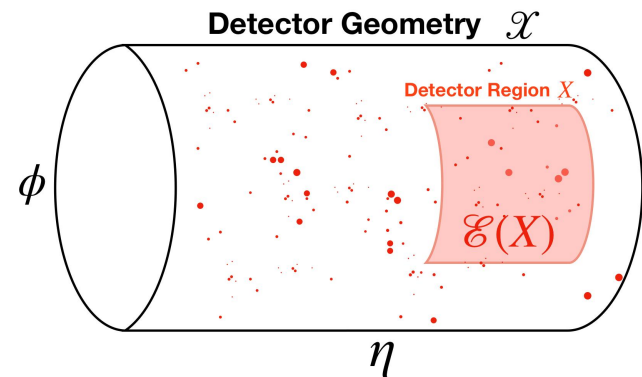
**In this talk, will treat “jets” and “events” as the same thing – the difference here is unimportant

Mathematical Details

Formally, an energy flow is a **measure** on the detector boundary

$$\mathcal{E}(X) = \int_X dx \mathcal{E}(x)$$

“How much total energy did I see in the detector region X ?”



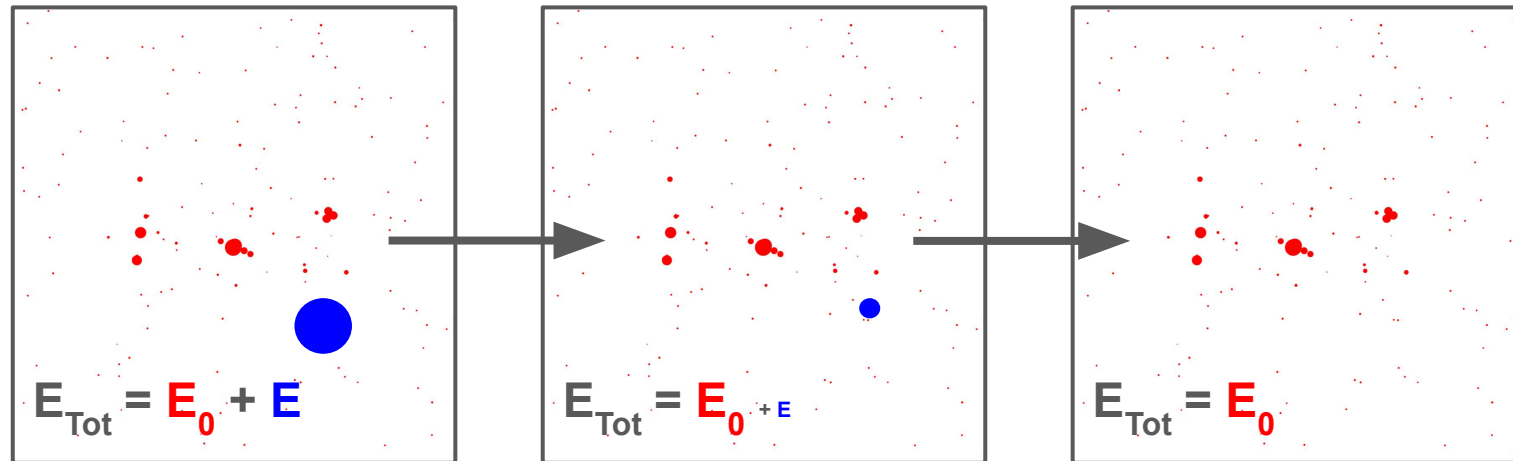
The fundamental operation on measures is **integration**:

$$\langle \mathcal{E}, \phi \rangle \equiv \int_{\mathcal{X}} dx \mathcal{E}(x) \phi(x)$$

“What is the energy-weighted expectation value of ϕ ?”

Mathematical Details – Topology

Definition (The **Weak* Topology**): A sequence of measures converges if all of their expectation values converge, as real numbers.

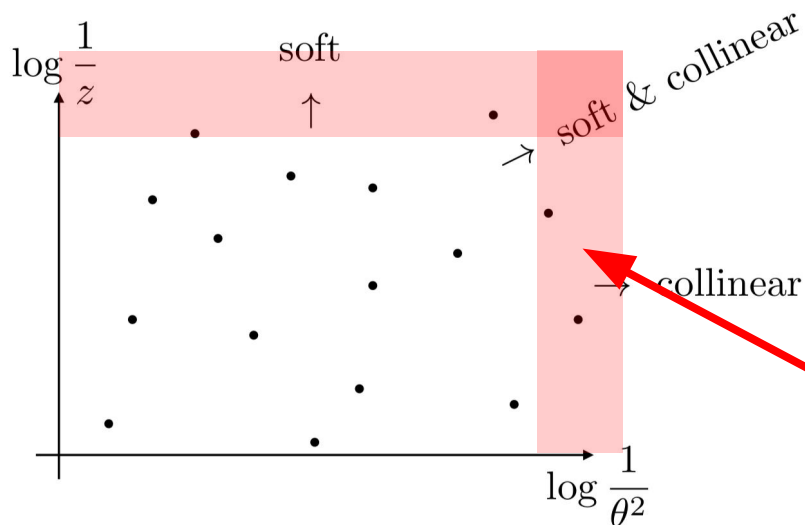


Definition (**Weak* Continuity**): An observable $\mathcal{O}(\mathcal{E})$ is continuous with respect to energy flows if, for any sequence of measures $\mathcal{E}_n$ that converges to $\mathcal{E}$, the sequence of real numbers $\mathcal{O}(\mathcal{E}_n)$ converges to $\mathcal{O}(\mathcal{E})$.

Topology $\Leftrightarrow$ IRC-Safety

Two ways to change the expectation values of an energy flow:

1. Change a particle's energy slightly, or add a low-energy particle - **IR**
2. Move a particle's position slightly, or split particles in two - **C**



An observable $\mathcal{O}$ is continuous if it changes only slightly under the above perturbations.

The regions of phase space causing IRC divergences is suppressed — $\mathcal{O}$ is **IRC-Safe**!

Geometry and the Wasserstein Metric

When are two events (distributions) similar? We need a metric to compare!

Properties we want:

1. ... is non-negative, non-degenerate^{*}, symmetric and finite — is an *actual* metric
 2. ... is weak^{*} continuous (IRC-safe)
 3. ... lifts the detector metric **faithfully** — preserves geometry
- ↓ We will break this soon!
- ↑ Explained shortly!

Geometry and the Wasserstein Metric

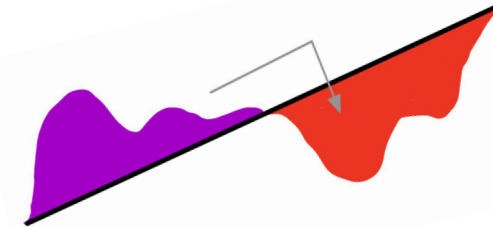
When are two events (distributions) similar? We need a metric to compare!

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EMD = Work done to move "dirt" optimally

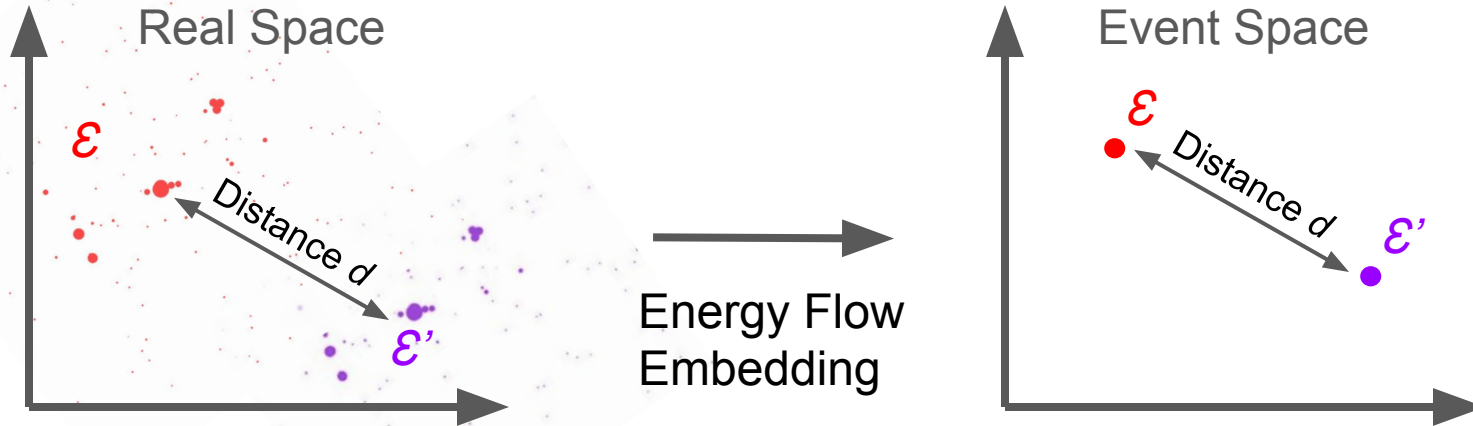
The only^{*} metric on distributions satisfying the above is the **Wasserstein Metric**:

$$\text{EMD}^{(\beta, R)}(\mathcal{E}, \mathcal{E}') = \min_{\pi \in \mathcal{M}(\mathcal{X} \times \mathcal{X})} \left[\frac{1}{\beta R^\beta} \left\langle \pi, d(x, y)^\beta \right\rangle \right] + |\Delta E_{\text{tot}}|$$

$$\pi(\mathcal{X}, Y) \leq \mathcal{E}'(Y), \quad \pi(X, \mathcal{X}) \leq \mathcal{E}(X), \quad \pi(\mathcal{X}, \mathcal{X}) = \min(E_{\text{tot}}, E'_{\text{tot}})$$

*There exist other metrics on distributions that are faithful only for very specific real-space distance norms, but we want them all!

The Importance of Being **Faithful***



A metric on events is **faithful** if, whenever two otherwise identical events ϵ and ϵ' are separated in real space by a distance d , the distance between the events is also d . Or d to a constant power

Only the **Wasserstein Metric** does this! → Can use to build **event** and **jet shapes** (old and new)!

Faithfulness also ensures very nice **numerical properties**, including no vanishing or exploding gradients.

Are there **other Metrics?** Yes.

If we:

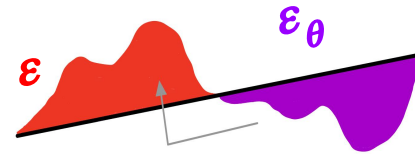
1. Only require *almost always a metric* — that is, only require $d(x,y) = 0$ *almost always* implies $x = y$.
2. Replace *faithfulness* with *isometry preserving* — that is $d(x,y) = 0$ if x and y are related by a translation, rotation, or mirror

Then we can write another Wasserstein-Based metric, called the **Spectral EMD**

$$\begin{aligned} \text{SEMD}_{\beta,p=2}(s_A, s_B) = & \sum_{i < j \in \mathcal{E}_A} 2E_i E_j \omega_{ij}^2 + \sum_{i < j \in \mathcal{E}_B} 2E_i E_j \omega_{ij}^2 \\ & - 2 \sum_{n \in \mathcal{E}_A^2, l \in \mathcal{E}_B^2} \omega_n \omega_l \left(\min [S_A(\omega_n^+), S_B(\omega_l^+)] - \max [S_A(\omega_n^-), S_B(\omega_l^-)] \right) \\ & \times \Theta (S_A(\omega_n^+) - S_B(\omega_l^-)) \Theta (S_B(\omega_l^+) - S_A(\omega_n^-)) , \end{aligned}$$

Unlike the EMD, the SEMD is very easy to *evaluate exactly and quickly!*

Technical Details ...



For $p = 2$, possible problem on the sp

For events A, B , the p **spectral EMD** is defined as (1D OT!):

EMD = Work done to move "dirt" optimally

$$\text{SEMD}_{\beta,p}(s_A, s_B) \equiv \int_0^{E_{\text{tot}}^2} dE^2 |S_A^{-1}(E^2) - S_B^{-1}(E^2)|^p$$

$$\begin{aligned} \text{SEMD}_{\beta,p=2}(s_A, s_B) = & \sum_{i < j \in \mathcal{E}_A} 2E_i E_j \omega_{ij}^2 + \sum_{i < j \in \mathcal{E}_B} 2E_i E_j \omega_{ij}^2 \\ & - 2 \sum_{n \in \mathcal{E}_A^2, l \in \mathcal{E}_B^2} \omega_n \omega_l (\min [S_A(\omega_n^+), S_B(\omega_l^+)] - \max [S_A(\omega_n^-), S_B(\omega_l^-)]) \\ & \times \Theta(S_A(\omega_n^+) - S_B(\omega_l^-)) \Theta(S_B(\omega_l^+) - S_A(\omega_n^-)), \end{aligned}$$

S = **cumulative spectral function**

$\pm$ indicates whether or not to include ω in the sum

The trick: Sum over pairs n of particles within each event.

Looks like $O(N^4)$, but with clever sorting & indexing in 1D, reduces to **$O(N^2)$** !

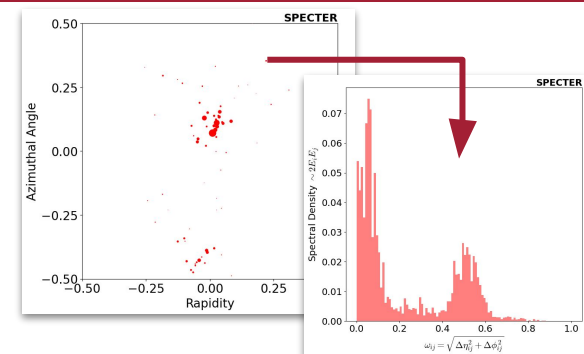
The **spectral density function**

$$s(\omega) = \sum_{i=1}^N \sum_{j=1}^N E_i E_j \delta(\omega - \omega(\hat{n}_i, \hat{n}_j))$$

Pairwise Distances

Reduces events to 1D, while preserving all* information about the event, up to translations and rotations.

*up to measure 0, but important degeneracies, ask me about this later!



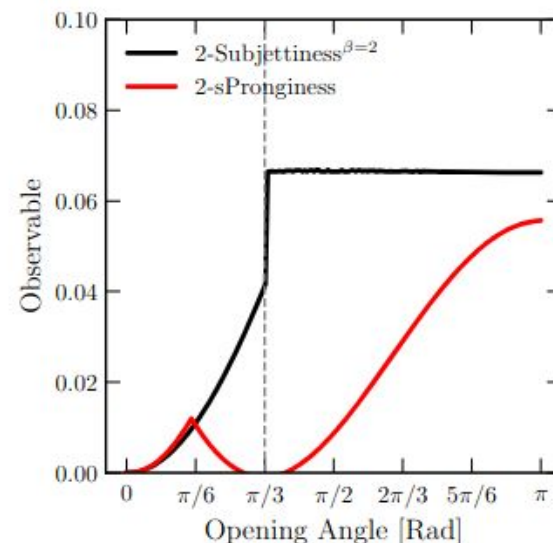
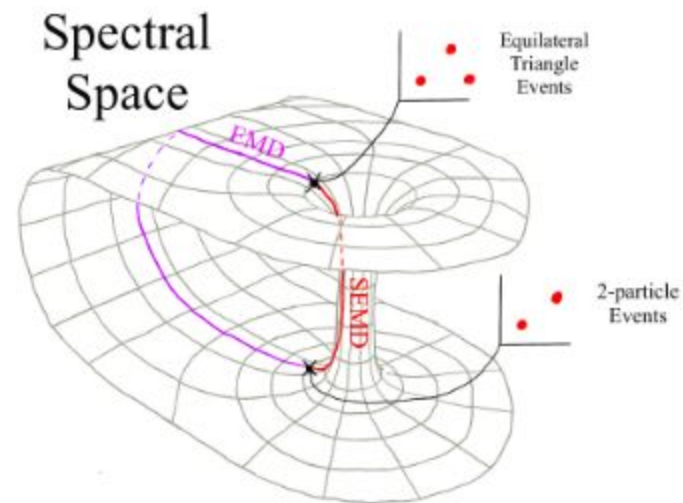
EMDs, ...

SEMD and Topology

The SEMD and EMD agree *locally*, but because the SEMD has measure-0 degeneracies, they can differ wildly for some events.

Two events far apart in EMD space can be extremely close in SEMD space, which changes the topology of the space.

However, the IRC structure is the same.



Part I

Can you Hear the **Shape** of a Jet?



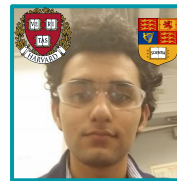
Download
SHAPER!

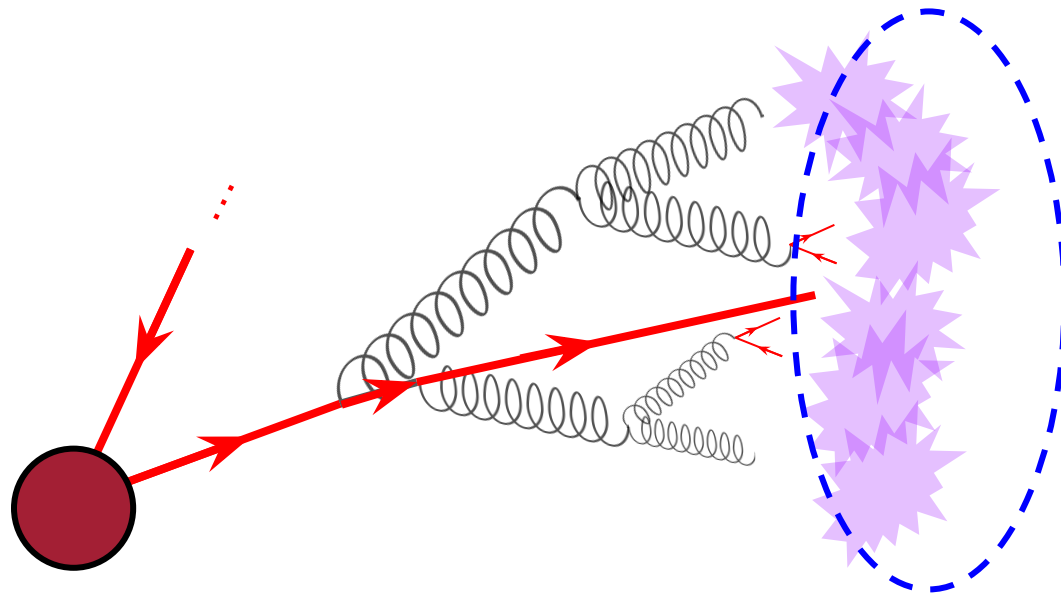


Download
SPECTER!

Try `pip install pyshaper`

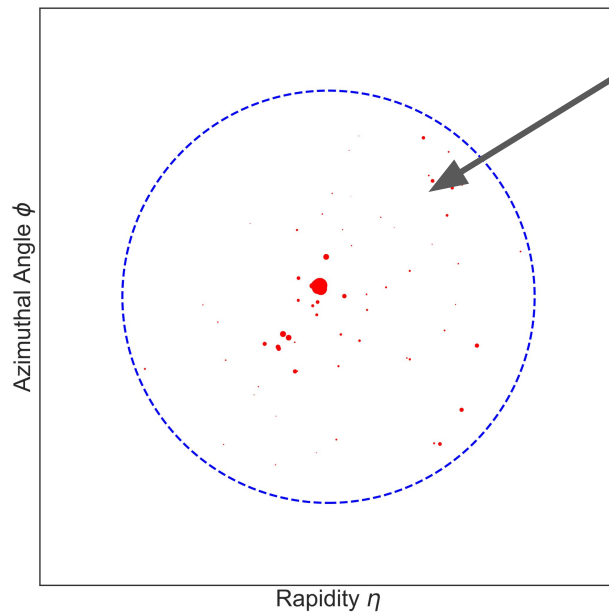
(New!) `pip install specterpy`





How “*big*” is this jet?

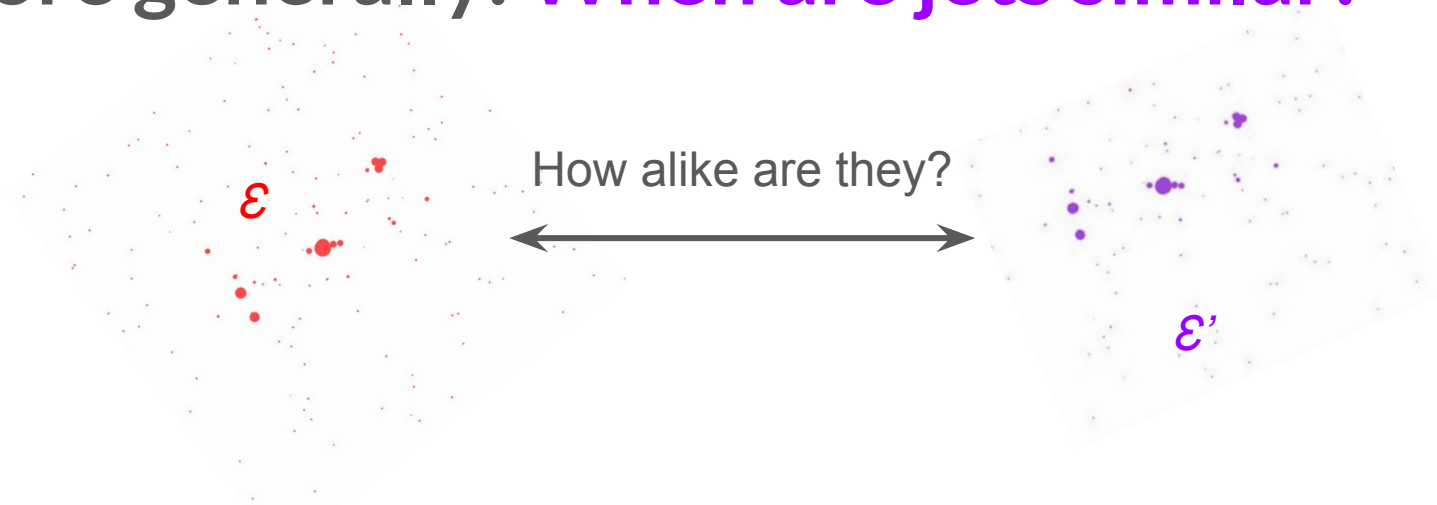
... What do we even mean by the “size” of a jet?



By following this line of thinking, we'll be able to come up with a new class of QCD observables!

I don't mean the fixed size of a jet algorithm, e.g. AK4, but rather a dynamical notion of a jet size!

More generally: When are jets similar?

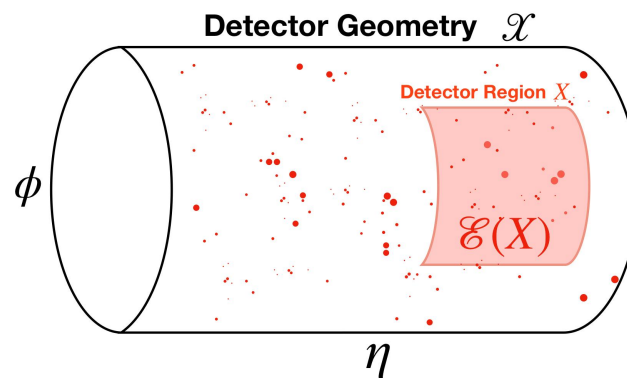


Jets can be represented as **point clouds** – in the last section we just saw two metrics that can be used to compare point clouds with nice properties!

$$\mathcal{E}(y) = \sum_i z_i \delta(y - y_i)$$

Detector Coordinate (η, ϕ)

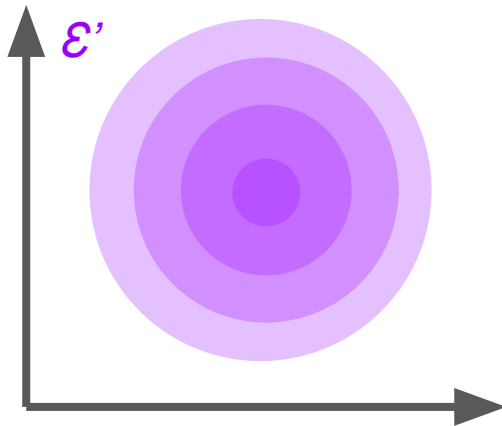
Energy Fraction E_i / E_{Tot}



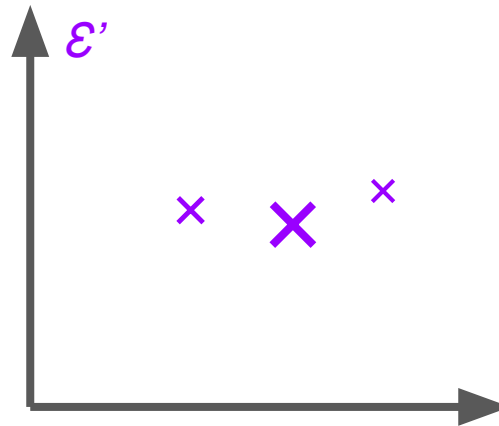
Shapes as Energy Flows

Energy flows don't have to be real events – they can be *any* energy distribution in detector space, or **shape**. In our framework, these shapes will later become our **Function Space**

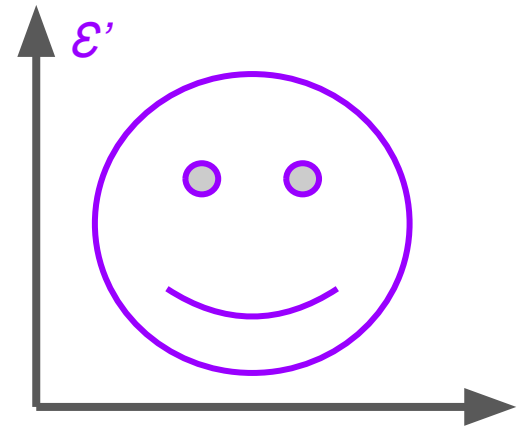
Can make anything you want! Even continuous or complicated shapes.
(Or, something you calculate in perturbative QCD)



Shape = 2D Gaussian



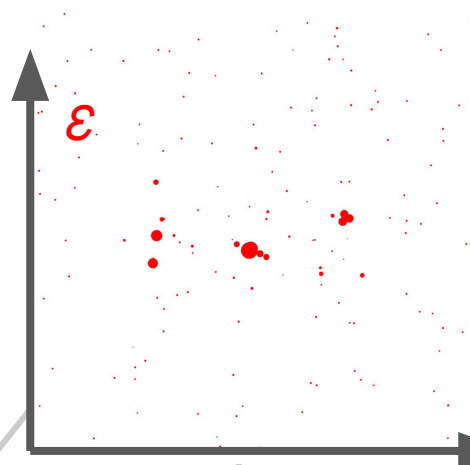
Shape = 3 Points



Shape = Smile

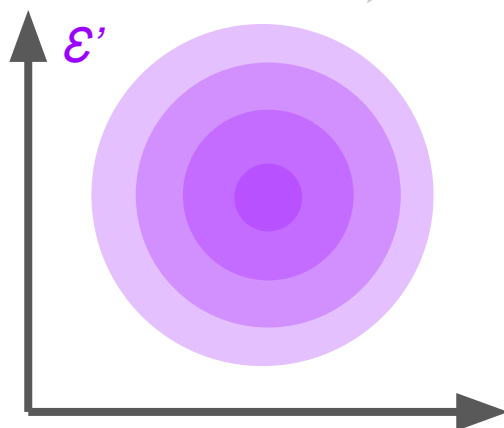
Shapiness

The EMD between a real event or jet $\mathcal{E}$ and idealized shape $\mathcal{E}'$ is the [shape]iness of $\mathcal{E}$ – a well defined IRC-safe observable!



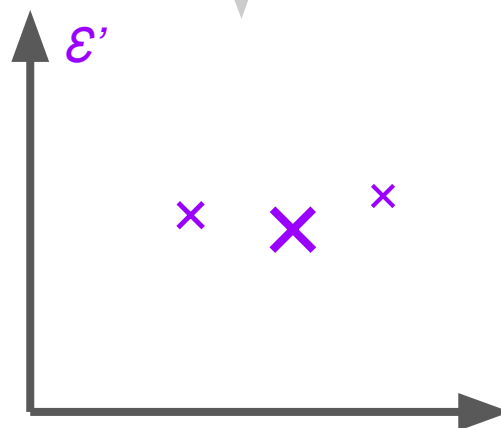
Answers the question:
“How much like the
shape $\mathcal{E}'$ is my event $\mathcal{E}$?”

Med EMD($\mathcal{E}$, $\mathcal{E}'$)
“Gaussiness”



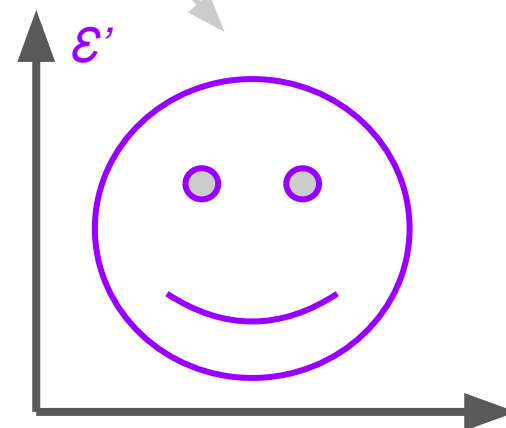
Shape = 2D Gaussian

Low EMD($\mathcal{E}$, $\mathcal{E}'$)
“3-Pointiness”
AKA “3-Subjettiness”



Shape = 3 Points

High EMD($\mathcal{E}$, $\mathcal{E}'$)
“Smileyiness”



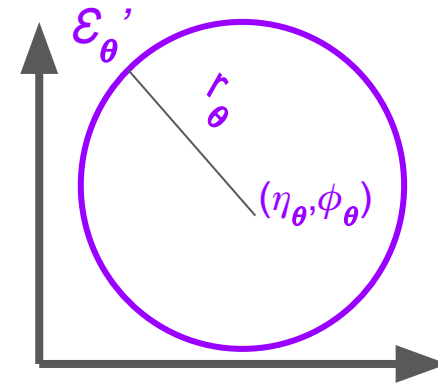
Shape = Smile

Mathematical Details - Shapiness

Rather than a single shape, consider a **parameterized manifold** $\mathcal{M}$ of **energy flows**.

e.g. The manifold of uniform circle energy flows:

$$\mathcal{E}_\theta'(y) = \begin{cases} \frac{1}{2\pi r_\theta} & |\vec{y} - \vec{y}_\theta| = r_\theta \\ 0 & |\vec{y} - \vec{y}_\theta| \neq r_\theta \end{cases}$$



Then, for an event $\mathcal{E}$, define the **shapiness** $\mathcal{O}_{\mathcal{M}}$ and **shape parameters** $\theta_{\mathcal{M}}$, given by:

$$\mathcal{O}_{\mathcal{M}}(\mathcal{E}) \equiv \min_{\mathcal{E}_\theta \in \mathcal{M}} \text{EMD}^{(\beta, R)}(\mathcal{E}, \mathcal{E}_\theta)$$

$$\theta_{\mathcal{M}}(\mathcal{E}) \equiv \underset{\mathcal{E}_\theta \in \mathcal{M}}{\text{argmin}} \text{EMD}^{(\beta, R)}(\mathcal{E}, \mathcal{E}_\theta)$$

Similar to a GAN, but we have only a few parameters

Observables $\Leftrightarrow$ Manifolds of Shapes

Observables can be specified solely by defining $\mathcal{O}_{\mathcal{M}}(\mathcal{E}) \equiv \min_{\mathcal{E}_{\theta} \in \mathcal{M}} \text{EMD}^{(\beta, R)}(\mathcal{E}, \mathcal{E}_{\theta})$,

a **manifold of shapes**:

$$\theta_{\mathcal{M}}(\mathcal{E}) \equiv \operatorname{argmin}_{\mathcal{E}_{\theta} \in \mathcal{M}} \text{EMD}^{(\beta, R)}(\mathcal{E}, \mathcal{E}_{\theta}),$$

Functional Minimization! Many well-known observables* already have this form!

Observable	Manifold of Shapes
N -Subjettiness	Manifold of N -point events
N -Jettiness	Manifold of N -point events with floating total energy
Thrust	Manifold of back-to-back point events
Event / Jet Isotropy	Manifold of the single uniform event

... and more!

All of the form “How much like **[shape]** does my **event** look like?”

Generalize to *any* shape.

*These observables are usually called event shapes or jet shapes in the literature – we are making this literal!

Hearing Jets Ring

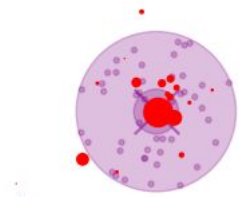
(and Disk, and Ellipse)

$\mathcal{O}_{\mathcal{M}}(\mathcal{E})$ answers: “How much like a **shape** in $\mathcal{M}$ does my **event** $\mathcal{E}$ look like?”

$\theta_{\mathcal{M}}(\mathcal{E})$ answers: “Which **shape** in $\mathcal{M}$ does my **event** $\mathcal{E}$ look like?”

Can define complex manifolds to probe increasingly subtle geometric structure!

Shape	Specification	Illustration
Ringiness $\mathcal{O}_R$	Manifold of Rings $\mathcal{E}_{x_0, R_0}(x) = \frac{1}{2\pi R_0}$ for $ x - x_0 = R_0$ $x_0 = \text{Center}, R_0 = \text{Radius}$	
Diskiness $\mathcal{O}_D$	Manifold of Disks $\mathcal{E}_{x_0, R_0}(x) = \frac{1}{\pi R_0^2}$ for $ x - x_0 \leq R_0$ $x_0 = \text{Center}, R_0 = \text{Radius}$	
Ellipsiness $\mathcal{O}_E$	Manifold of Ellipses $\mathcal{E}_{x_0, a, b, \varphi}(x) = \frac{1}{\pi ab}$ for $x \in \text{Ellipse}_{x_0, a, b, \varphi}$ $x_0 = \text{Center}, a, b = \text{Semi-axes}, \varphi = \text{Tilt}$	
(Ellipse Plus Point)iness	Composite Shape $\mathcal{O}_E \oplus \tau_1$ Fixed to same center x_0	
N-(Ellipse Plus Point)iness Plus Pileup	Composite Shape $N \times (\mathcal{O}_E \oplus \tau_1) \oplus \mathcal{I}$	



Disk + δ -function



Ring + δ -function

Probing **collinear** (δ -function) and **soft** (shape) structure!

Some examples of new shapes you can define!

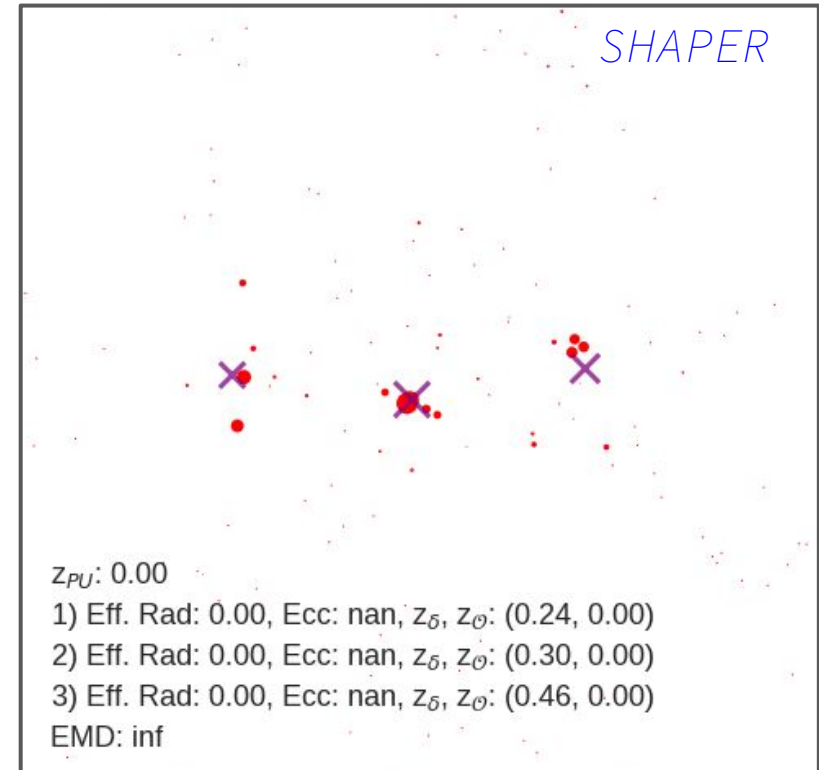
New IRC-Safe Observables

The *SHAPER* framework makes it easy to algorithmically invent new jet observables!

e.g. *N-(Ellipse+Point)iness+Pileup* as a jet algorithm:

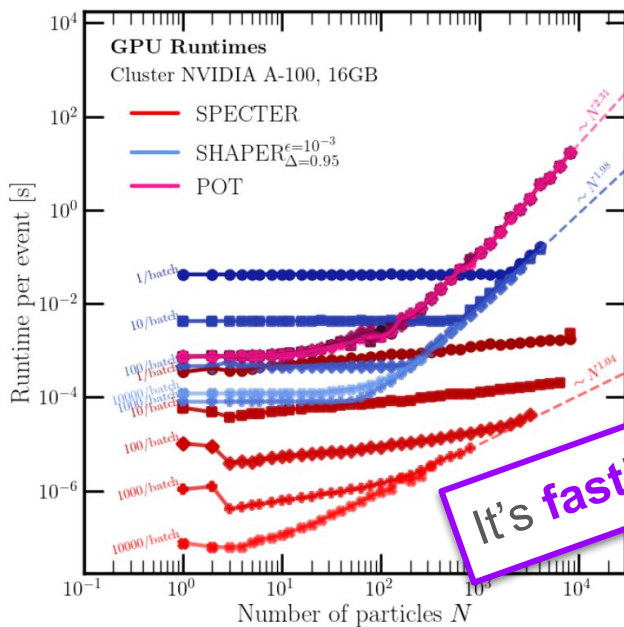
- Learn jet centers + collinear radiation
- Dynamic jet radii (no R parameter!)
- Dynamic eccentricities and angles
- Dynamic jet energies
- Dynamic pileup (no z_{cut} parameter!!!)
- Learned parameters for discrimination

Can design custom specialized jet algorithms to learn jet substructure!

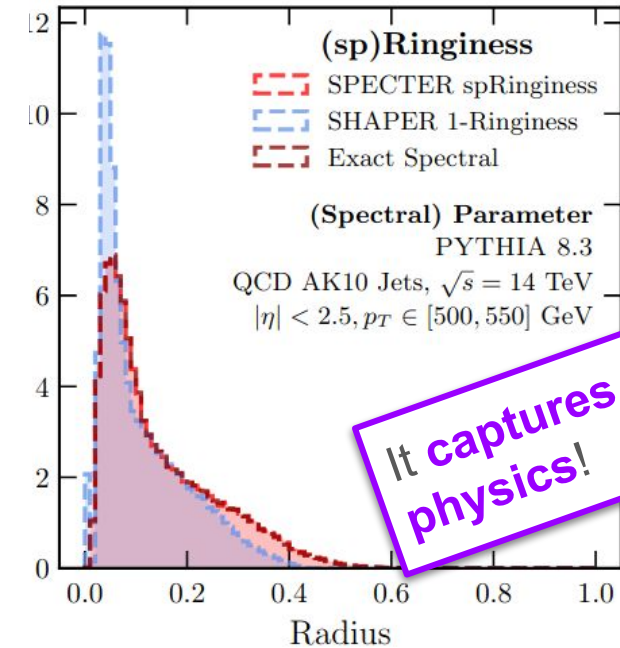
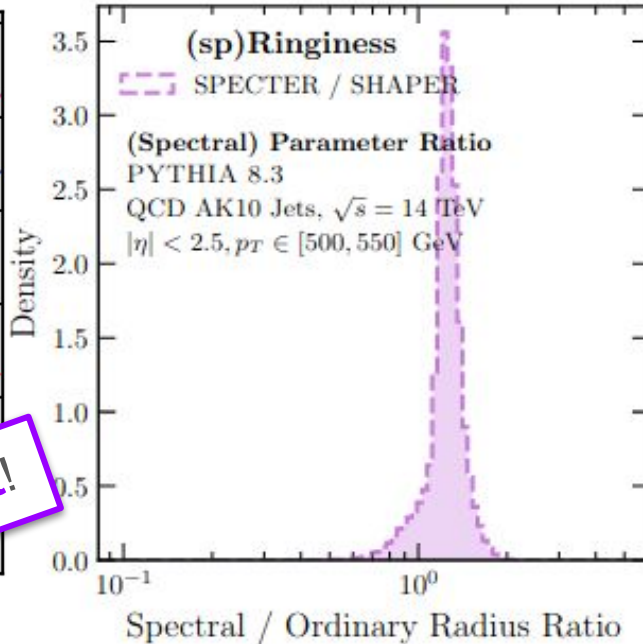


SPECTER: OT, but *FAST*

By demanding our loss is only *almost** always a metric on events rather than always a metric, we can build a new loss that is *completely and exactly solvable*!



It's fast!



It captures physics!



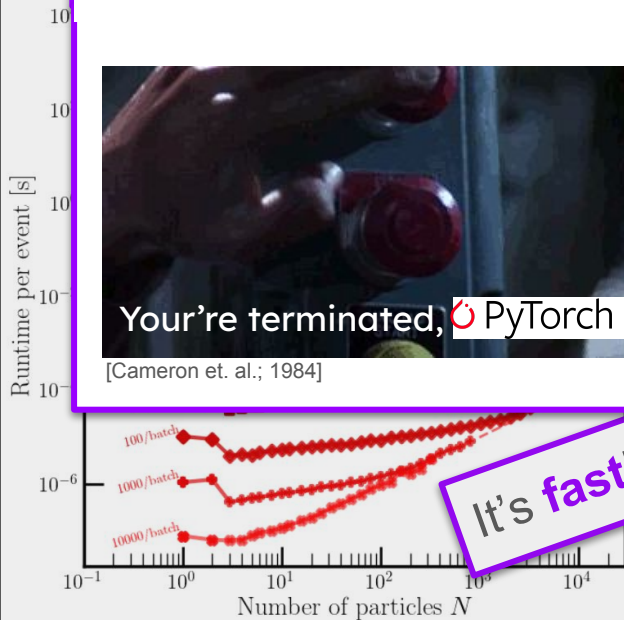
SPECTER: OT, but FAST

Exact and closed form
Spectral Jet Radius!
No gradient descent or
approximations required!

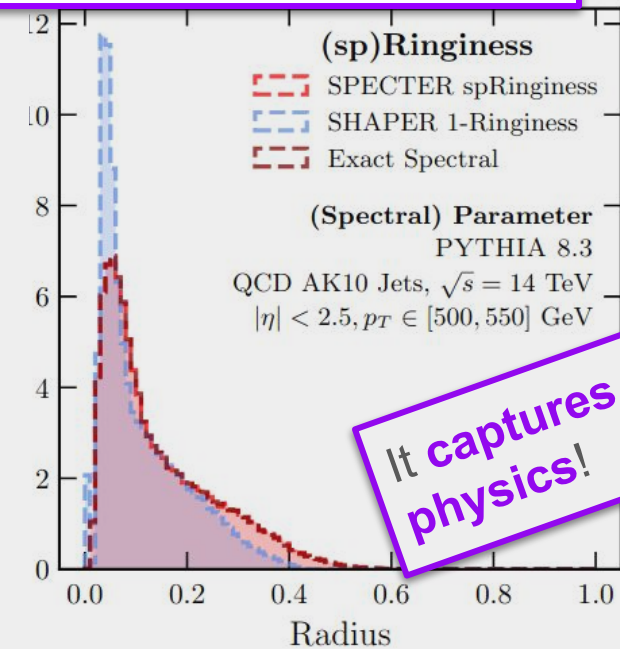
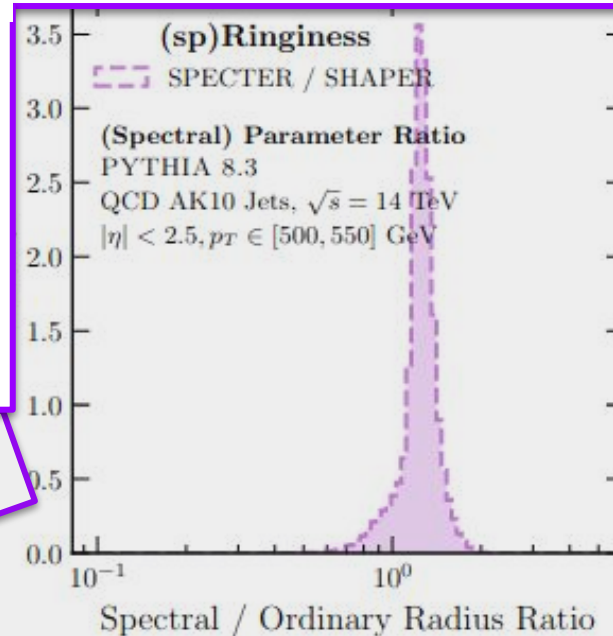
$$\text{opt} = \frac{2}{\pi} \sum_{\substack{n \in \mathcal{E}^2 \\ \omega_n < \omega_{n+1}}} \omega_n \left[\cos \left(\frac{\pi}{2E_{\text{tot}}^2} S^-(\omega_n) \right) - \cos \left(\frac{\pi}{2E_{\text{tot}}^2} S^+(\omega_n) \right) \right]$$



[Cameron et. al.; 1984]



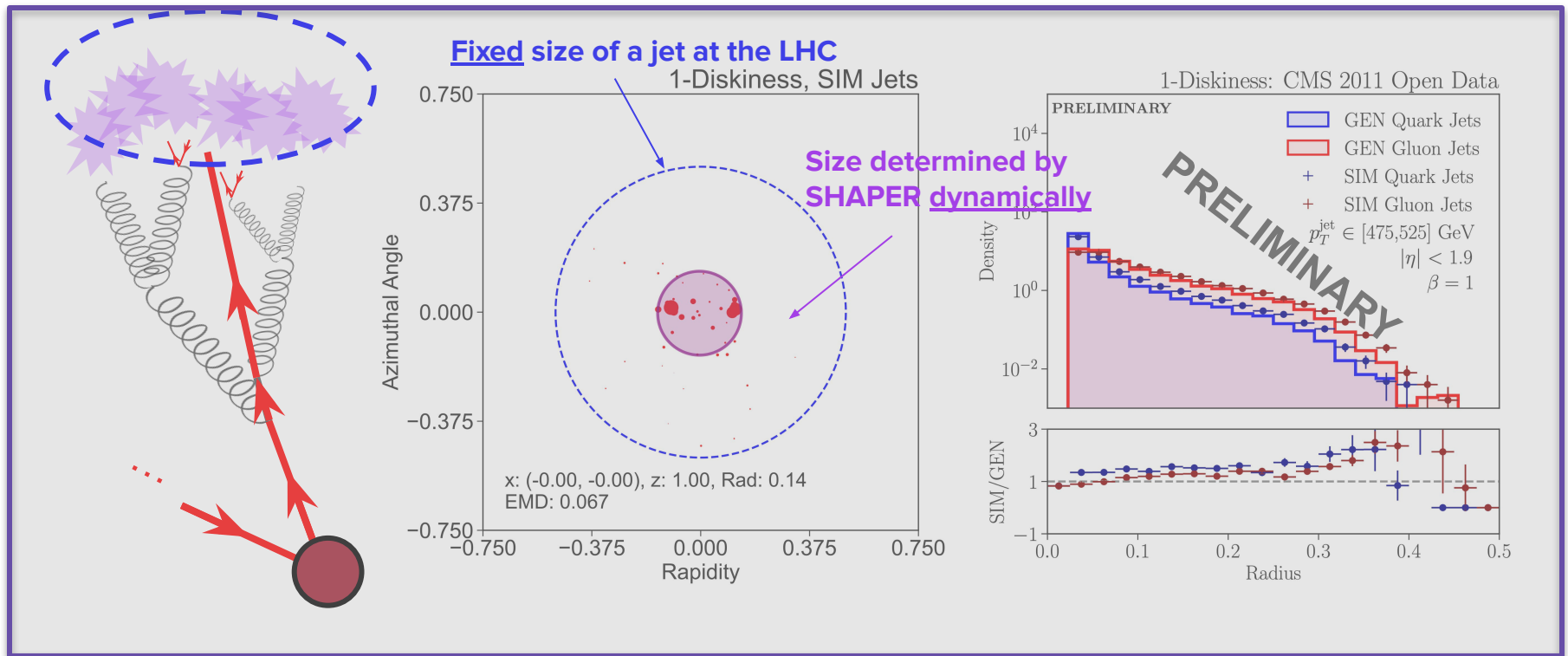
It's fast!



It captures physics!

Back to our question: How Big are Jets?

Slide images from [Wu; APS 2023]



We can now answer this question in a precise way!



What else? Heavy Ion Jets

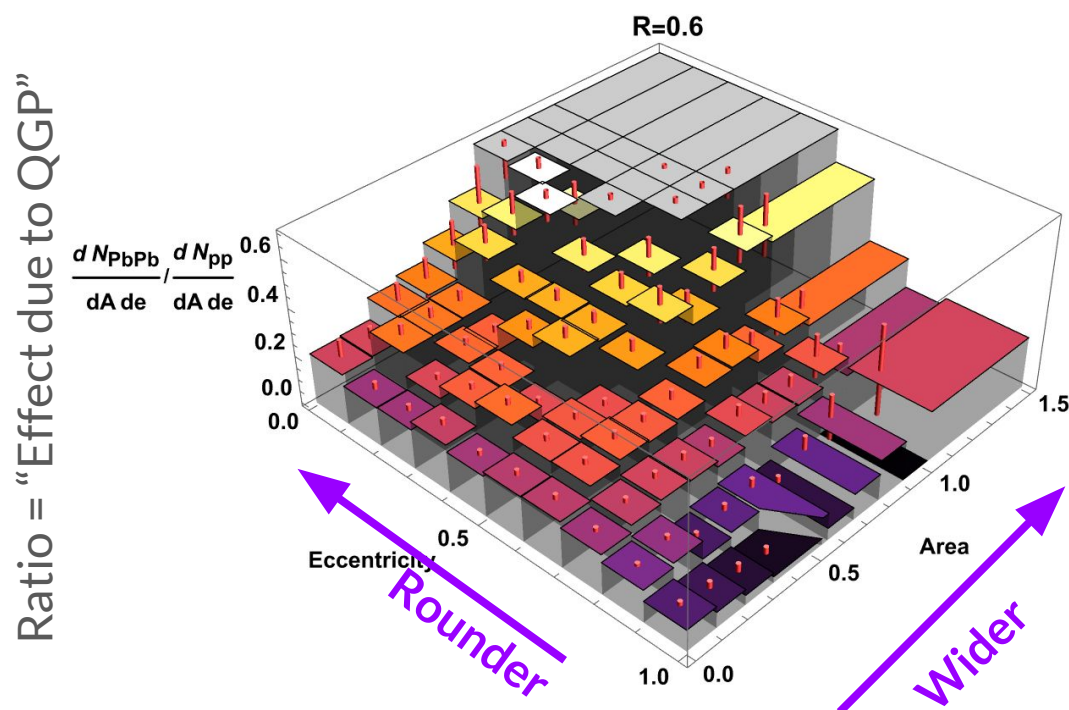
Exploring the shape of jets in Heavy Ion Collisions

The quark-gluon plasma is expected to change the (distribution of the) shape of jets.

The survival rate depends on both the *area* of the jet and the *eccentricity*.

We can directly see that the medium makes jets *wider* and *rounder*.

Fitting **ellipses*** to jets in **Pb-Pb** and **pp**



*These ellipses are not uniform, but rather have exponential falloff from the center

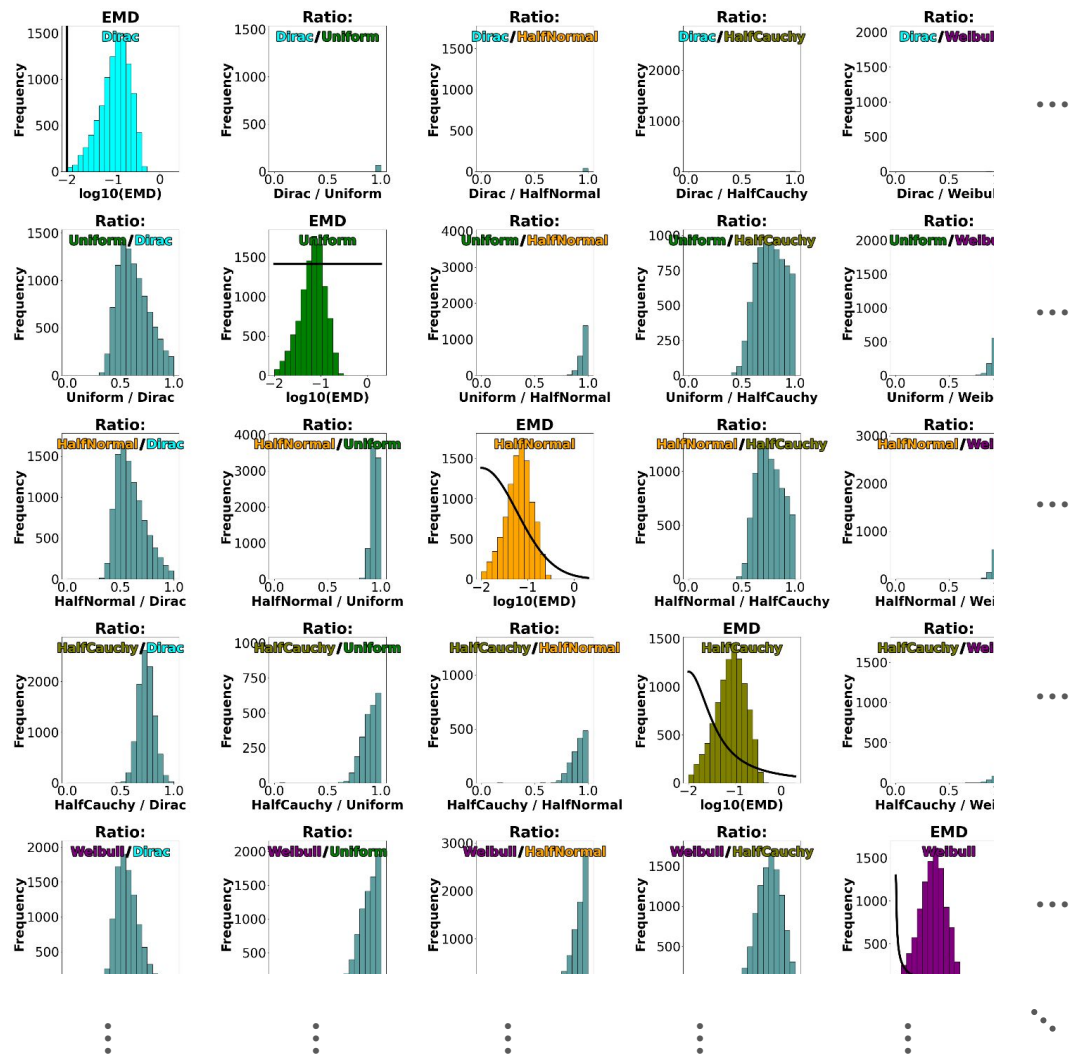
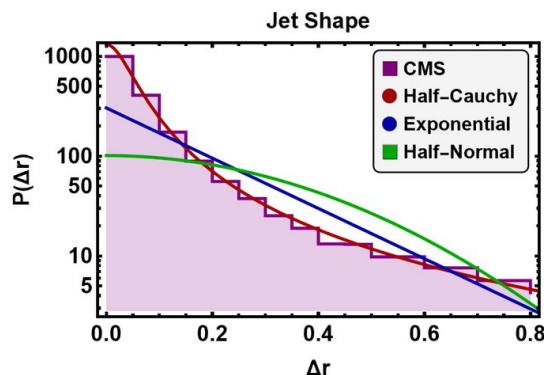
What else? Heavy Ion Jet Profiles

What is the *best* jet shape?

Important distinction: Shape of the *average* jet versus *average* shape of a jet (plotted)

No single simple distribution can capture all jets.

The shape of the *average* jet is close to half-Cauchy, even though any given jet is not!

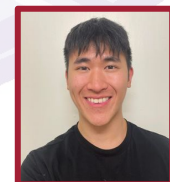


Part II

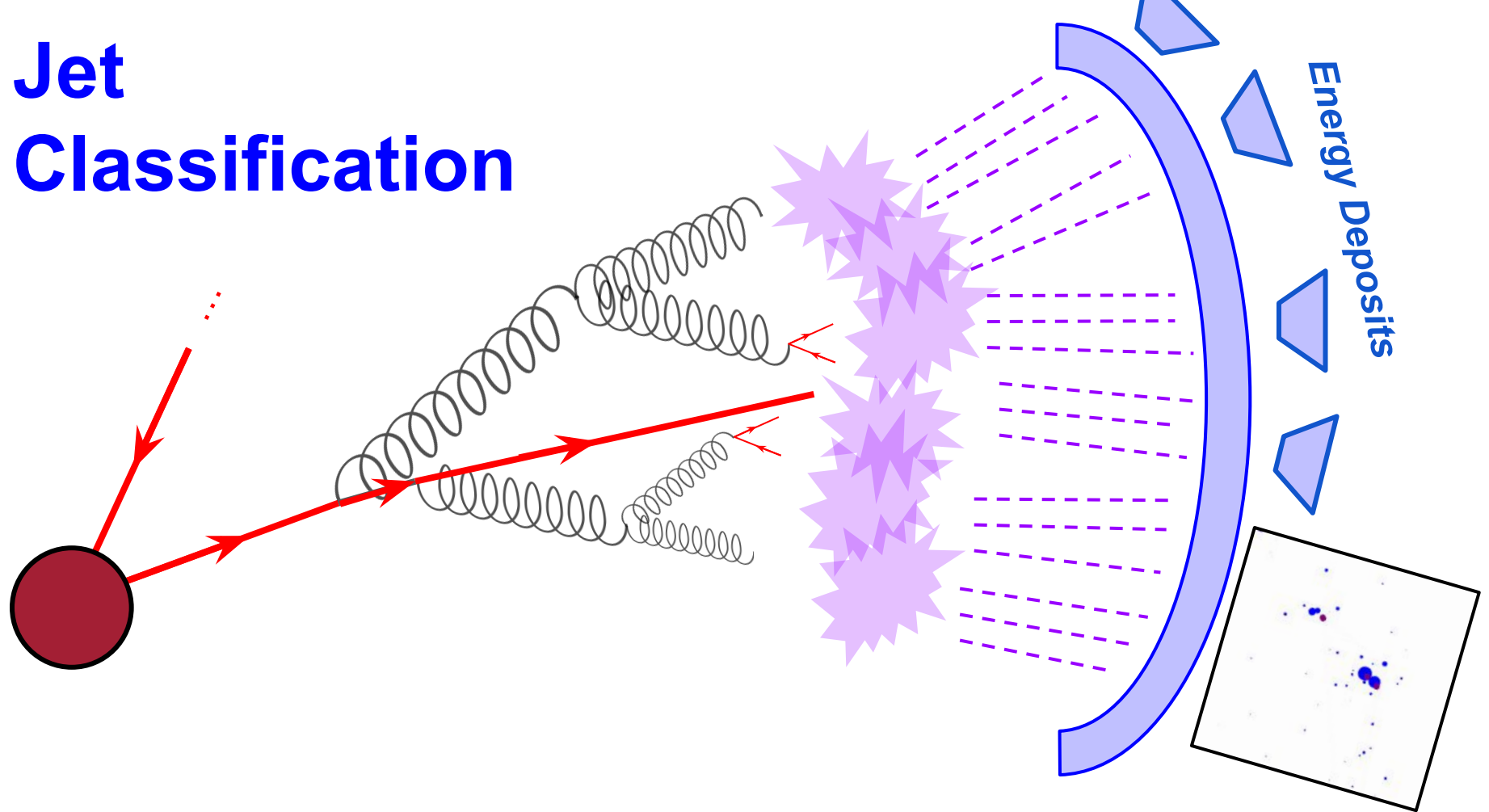
Moments of Clarity: Streamlining Latent Spaces



Download
Moment EFNs!



Jet Classification

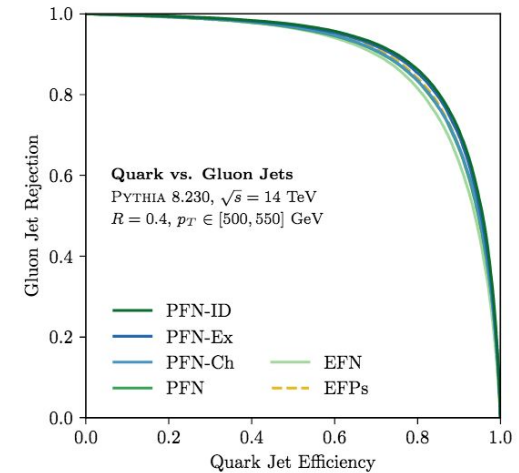
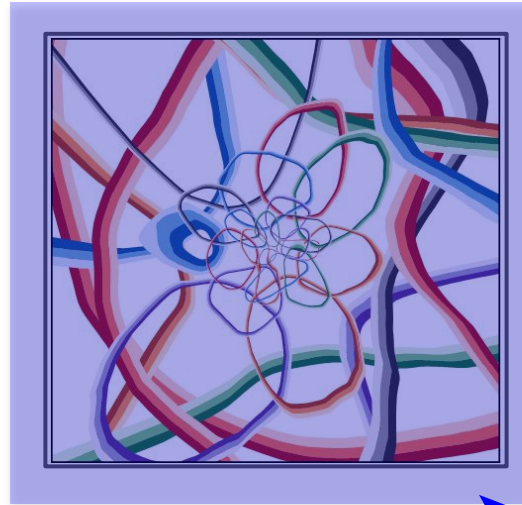
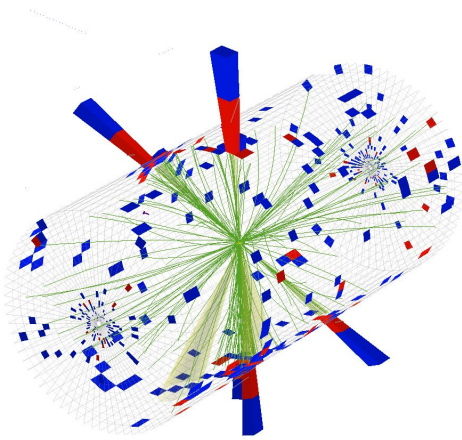


Given the final measured **point cloud**, was the **initiating parton** a **quark** or a **gluon**?

A staple machine learning task!

Rikab Gambhir – LBNL – 04 March 2025

Typical Machine Learning Setup



Collider Data
 ~1000 Dimensional

Latent Space
 ~10-100 Dimensional

Observables
 ~1-10 Dimensional

Pictured: An Energy Flow Network (**EFN**):

$$\mathcal{O}(\{p_1, \dots, p_M\}) = F \left(\sum_{i=1}^M z_i \Phi(\hat{p}_i) \right)$$

(How) can we understand and constrain this?

(How) can we be more efficient?

Deep Learning on Point Clouds

The **Deep Sets Theorem** tells us how to parameterize functions on point clouds*:

Set of momenta $\rightarrow$ $a = 1 \dots L$, the *Latent Dimension* index

$$\mathcal{O}(\mathcal{P}) = F(\langle \phi^a \rangle_{\mathcal{P}})$$

The Deep Sets Theorem guarantees that any function on sets $\mathcal{P}$ can be written this way, for “sufficiently complex” F and ϕ and large enough L .

$\langle \phi \rangle_{\mathcal{P}} \equiv \sum_i z_i \phi(\hat{p}_i)$

For this talk, I will focus primarily on Energy Flow Networks (EFNs) where the sums are energy-weighted.

*More sophisticated architectures still reduce to Deep Sets in some limit, e.g. graph networks

Deep Learning on Point Clouds

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$\langle \phi \rangle_{\mathcal{P}} \equiv \sum_i z_i \phi(\hat{p}_i)$

But how complex do F and ϕ need to be? Can I reduce the number of latent dimensions?

For this talk, I will focus primarily on Energy Flow Networks (EFNs) where the sums are energy-weighted.

*More sophisticated architectures still reduce to Deep Sets in some limit, e.g. graph networks

Moment Pooling

$$\mathcal{O}(\mathcal{P}) = F(\langle \phi^a \rangle_{\mathcal{P}})$$



Generalize to more moments! “**Moment Pooling**”

$$\mathcal{O}_k(\mathcal{P}) = F_k(\underbrace{\langle \phi^a \rangle_{\mathcal{P}}}_{1^{\text{st}} \text{ Moment}}, \underbrace{\langle \phi^{a_1} \phi^{a_2} \rangle_{\mathcal{P}}}_{2^{\text{nd}} \text{ Moment}}, \dots, \underbrace{\langle \phi^{a_1} \dots \phi^{a_k} \rangle_{\mathcal{P}}}_{k^{\text{th}} \text{ Moment}})$$

$k = \text{Highest order moment considered}$

$$L_{\text{eff}} = \binom{L+k}{k}$$

Moment Pooling – Why?

k = Highest order moment considered ↓

$$\mathcal{O}_k(\mathcal{P}) = F_k \left(\underbrace{\langle \phi^a \rangle_{\mathcal{P}}}_{1^{\text{st}} \text{ Moment}}, \underbrace{\langle \phi^{a_1} \phi^{a_2} \rangle_{\mathcal{P}}}_{2^{\text{nd}} \text{ Moment}}, \dots, \underbrace{\langle \phi^{a_1} \dots \phi^{a_k} \rangle_{\mathcal{P}}}_{k^{\text{th}} \text{ Moment}} \right)$$

$L_{\text{eff}} = \binom{L+k}{k}$

- **Explicit Multiplication:** Neural nets are mostly piecewise linear! But most functions we learn aren't. Moments are just multiplication, but for distributions!
- **Latent Space Compression:** For large L , there are effectively L^k latent dimensions due to combinatorics, but still made of only L functions! Fewer latent dimensions means easier analysis!

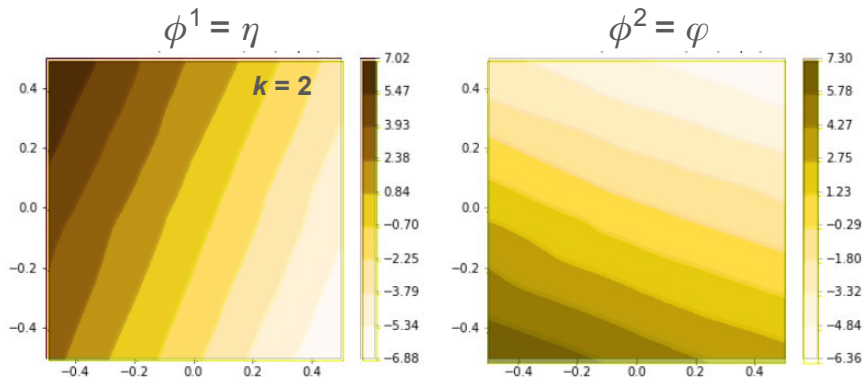
This is actually a much simpler version of the attention mechanism. It turns out **you don't even need attention** – ask me later!

Multiplication is all You Need

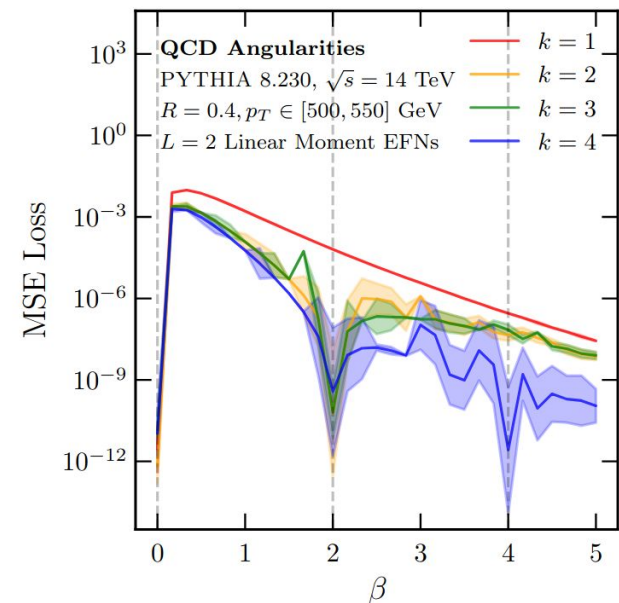
In the moment language, even integer^{*} β jet angularities $\Leftrightarrow k = \beta^{\text{th}}$ moments!

$$\begin{aligned}\lambda^{(\beta)}(\mathcal{P}) &= \sum_i z_i (\eta_i^2 + \phi_i^2)^{\beta/2} \\ &= \langle \eta^\beta \rangle_{\mathcal{P}} + \langle \phi^\beta \rangle_{\mathcal{P}} + \text{Cross Moments}\end{aligned}$$

Learns the simplest latent representations!

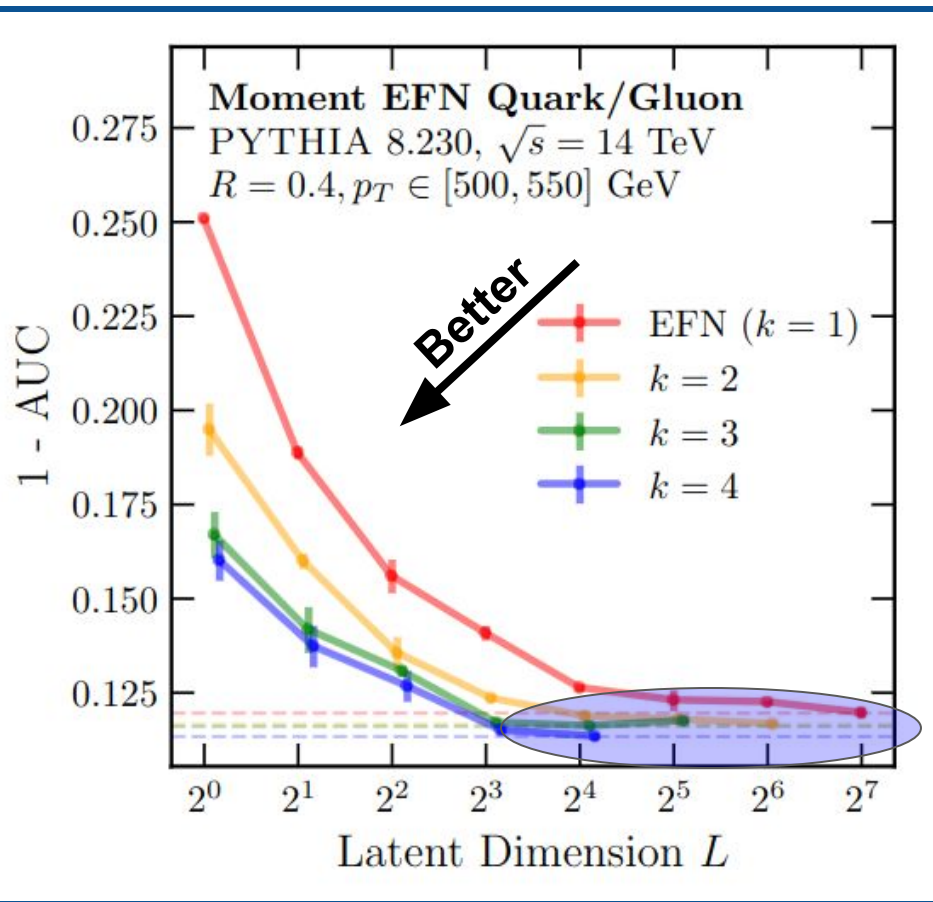


Encourage learning a diagonal basis in (η, ϕ) using L1 Regularization on F



The lesson: Basic observables are naturally represented with multiplication, so build it in!

Moment Pooling Results

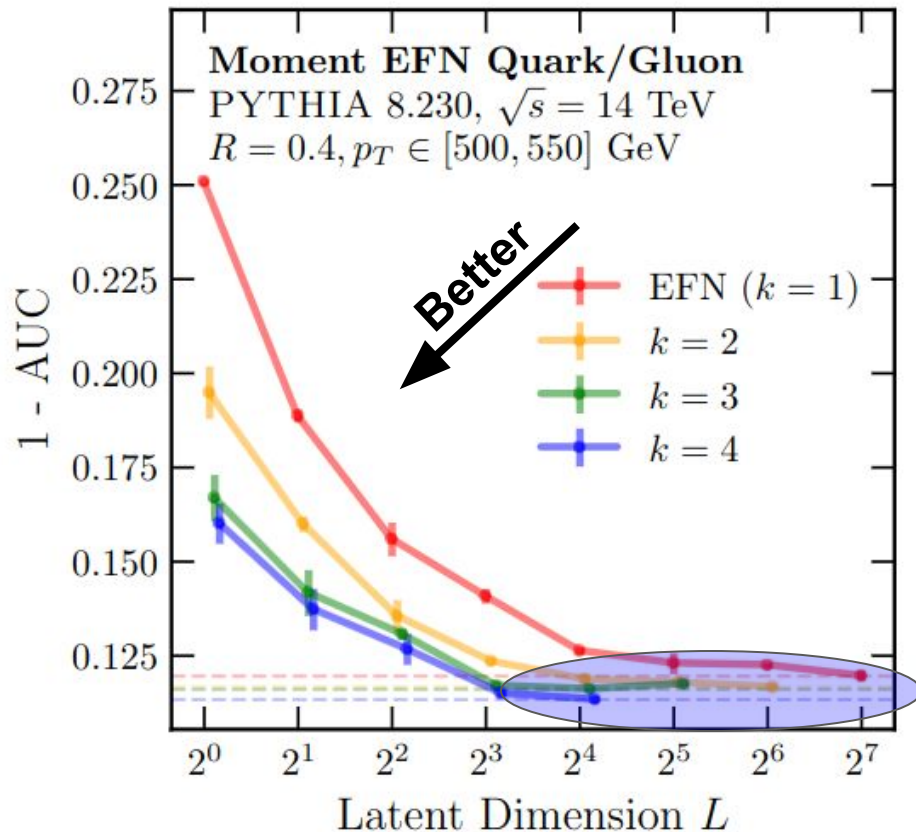


A high order **Moment EFN** can achieve the same top performance on quark/gluon classification as an **ordinary EFN**, but with far fewer latent dimensions!*

A Moment EFN is a more efficient encoding of the same data!

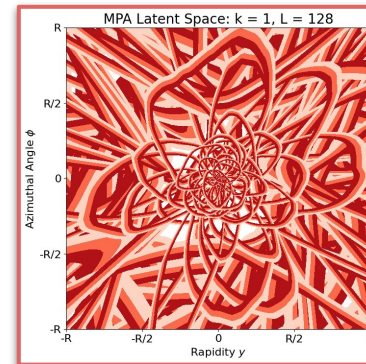
*Interestingly, the performance is constant in the *effective* latent dimensions and number of parameters. Ask me about this!

The Black Box

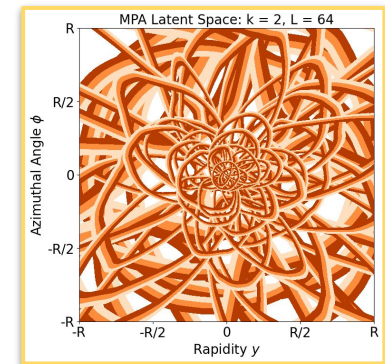


Latent Spaces

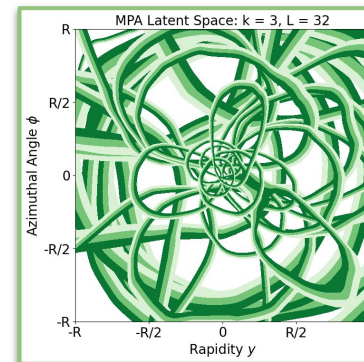
EFN ($k = 1$)
 $L = 128$



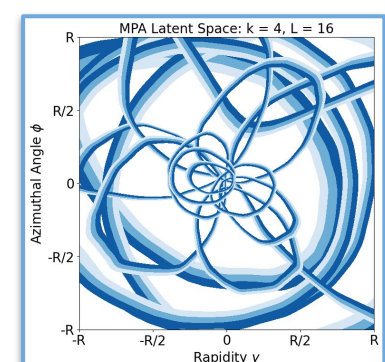
$k = 2$
 $L = 64$



$k = 3$
 $L = 32$



$k = 4$
 $L = 16$

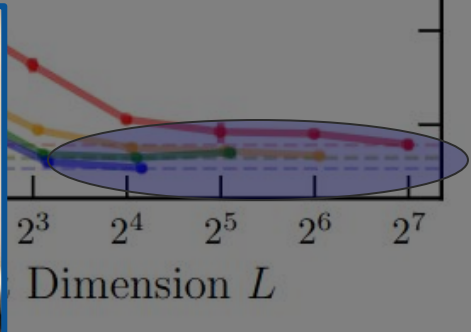


The 45% - 55% contours of each of the L different ϕ functions

*Interestingly, the performance is constant in the *effective* latent dimensions and number of parameters. Ask me about this!

So what? The information is just being shuffled around. Is 16 latent dimensions any more interpretable than 128?

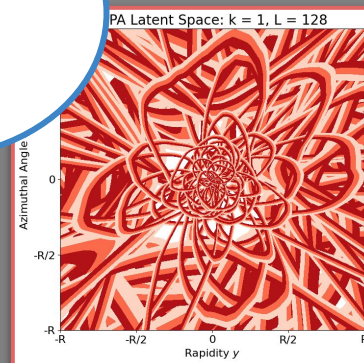
The machine probably learned something, but I didn't!



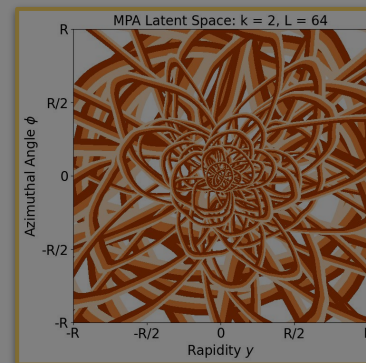
Me, just before the **moment** of clarity.

Latent Spaces

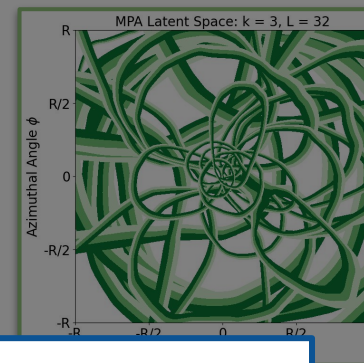
EFN ($k = 1$)
 $L = 128$



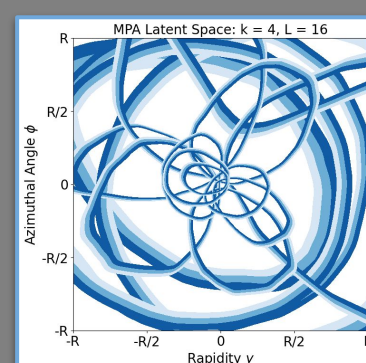
$k = 2$
 $L = 64$



$k = 3$
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$k = 4$
 $L = 16$



*Interestingly, the performance is constant in the *effective* latent dimensions and number of parameters. Ask me about this!

Latent Spaces

So what? The information is just being shuffled around. Is 16 latent dimensions any more interpretable than 128?

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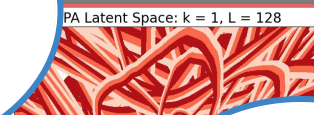
I'm allowed to control the ML function space!

Let me just choose one that's easy for me to understand: **One dimensional functions**

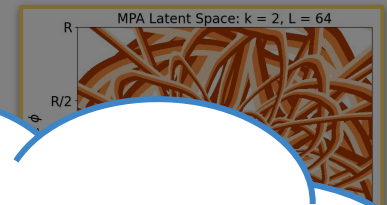
Because of latent space compression, even one dimension can contain a lot of info!

Me, just before the **moment** of clarity.

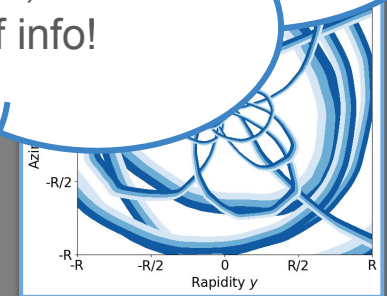
EFN ($k = 1$)
 $L = 128$



$k = 2$
 $L = 64$

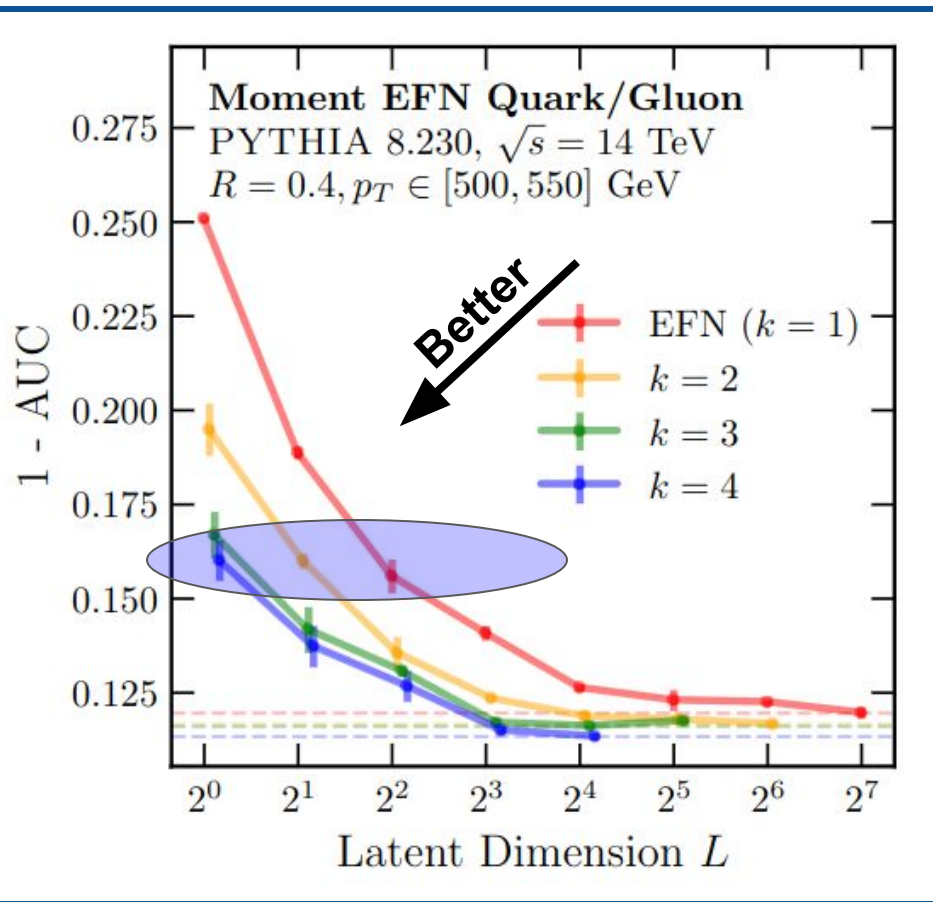


2^3 2^4 2^5
Dimension L



*Interestingly, the performance is constant in the *effective* latent dimensions and number of parameters. Ask me about this!

Moment Pooling Results (Cont'd)



Let's instead look at networks with just a *single* latent dimension.

A **$k = 4$ Moment EFN** with *just one* latent dimension does just as good as an **ordinary EFN** with *four* latent dimensions!

Going down to $L = 1$

The more efficient encoding is extremely useful when we can get to $L = 1$!
Look how much simpler allowing neural networks to multiply makes things!

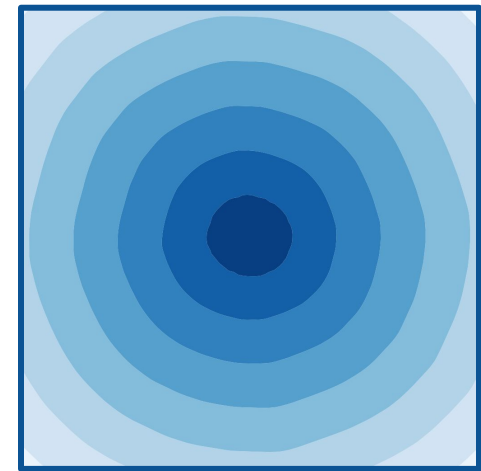
EFN ($k = 1$)
 $L = 4$



Same information!



Order $k = 4$
 $L = 1$



- 4 Different Functions
- Combined in a complicated way
- No symmetry
- ???

- Single function
- Expected to be, and is, radially symmetric
- We can **interpret** this!

The Moment of Truth

We can extract physics from this!

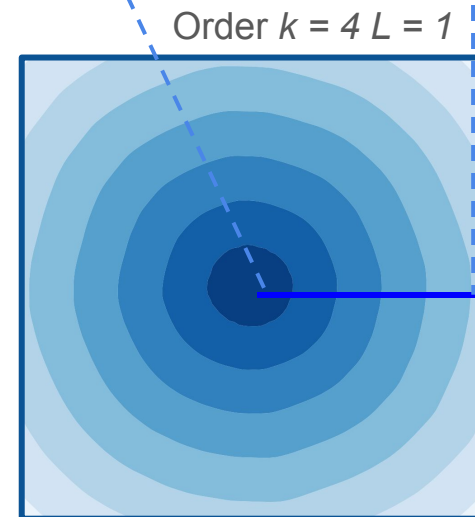
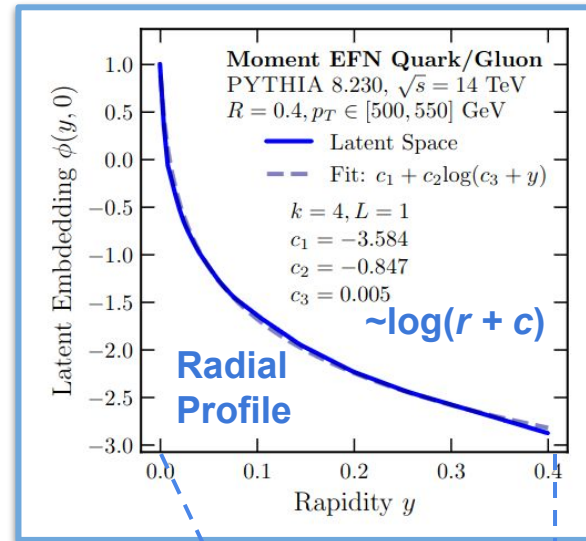
$$\Phi_{\mathcal{L}}(r) = c_1 + c_2 \log(c_3 + r)$$

This defines a **log angularity** observable, related to the $\square \rightarrow 0$ limit of ordinary angularities

We can even see non-perturbative physics!

$$c_3 \sim \frac{\Lambda_{\text{QCD}}}{p_T R} \sim 0.001$$

Just this alone is enough to get an IRC-safe quark-gluon classifier with an AUC ~ 0.83 !



Part III: Re-Normalizing Flows

Taming Logs and Perturbation Theory in QCD

[**Work in Progress.** Suggestions and Comments very welcome!]



Perturbation Theory in QCD

Now that we've seen plenty of new observables x , let's try to actually calculate them!
We'll perturb in a small parameter α to calculate the distribution of x :

$$p(x|\alpha) = \underbrace{p_0(x) + \alpha p_1(x|\alpha) + \alpha^2 p_2(x|\alpha) + \dots + \alpha^M p_M(x|\alpha)}_{\text{Fixed-order calculation}} + \underbrace{\mathcal{O}(\alpha^{M+1})}_{\text{Unknown}}$$

Example: Thrust [Singular parts Becher, Schwartz; [0803.0342](#)]

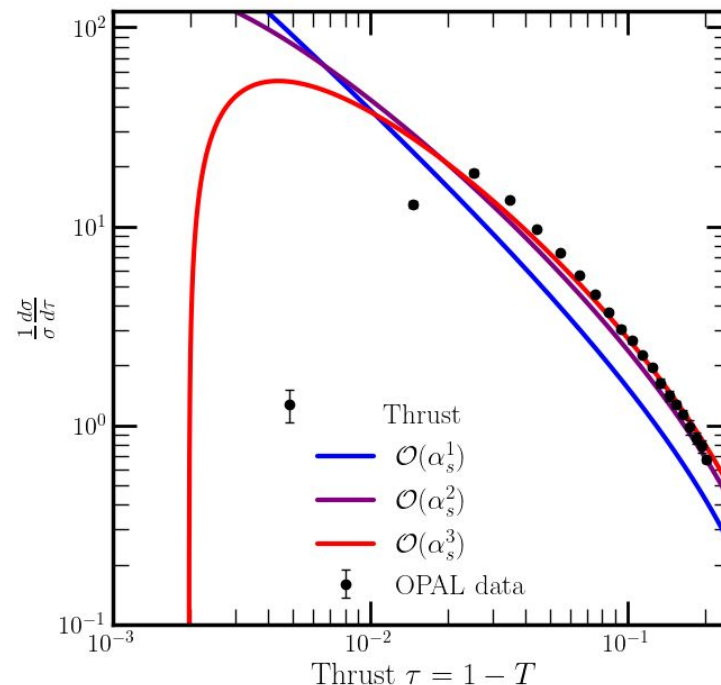
$$p_1 = \frac{1}{\tau} \{C_F [-4 \ln \tau - 3]\}$$

$$p_2 = \frac{1}{\tau} \left\{ C_F^2 \left[8 \ln^3 \tau + 18 \ln^2 \tau + (13 - 4\pi^2) \ln \tau + \frac{9}{4} - 2\pi^2 + 4\zeta_3 \right] \right. \\ \left. + C_F T_F n_f \left[-4 \ln^2 \tau + \frac{22}{9} \ln \tau + 5 \right] \right. \\ \left. + C_F C_A \left[11 \ln^2 \tau + \left(-\frac{169}{18} + \frac{2\pi^2}{3} \right) \ln \tau - \frac{57}{4} + 6\zeta_3 \right] \right\},$$

$$p_3 = \frac{1}{\tau} \left\{ C_F^3 \left[-8 \ln^5 \tau - 30 \ln^4 \tau + \ln^3 \tau \left(-44 + \frac{40\pi^2}{3} \right) + \ln^2 \tau (-88\zeta_3 + 24\pi^2 \right. \right. \\ \left. \left. - 27) + \ln \tau \left(-c_{2C_F}^S - 96\zeta_3 - \frac{53\pi^4}{90} + \frac{79\pi^2}{6} - \frac{17}{2} \right) + 16\pi^2\zeta_3 - 39\zeta_3 - 132\zeta_5 \right] \right. \\ \left. + C_F^2 T_F n_f \left[-4 \ln^4 \tau + \frac{22}{3} \ln^3 \tau + \left(-\frac{169}{3} + \frac{2\pi^2}{3} \right) \ln^2 \tau + \left(-\frac{169}{3} + \frac{2\pi^2}{3} \right) \ln \tau + \frac{9}{4} - 2\pi^2 + 4\zeta_3 \right] \right. \\ \left. + C_F T_F^2 n_f^2 \left[-4 \ln^3 \tau + \frac{22}{3} \ln^2 \tau + \left(-\frac{169}{3} + \frac{2\pi^2}{3} \right) \ln \tau + \frac{9}{4} - 2\pi^2 + 4\zeta_3 \right] \right\},$$

... [many more terms]

For comparison: **OPAL** Data



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$$p(x|\alpha) = \underbrace{p_0(x) + \alpha p_1(x|\alpha) + \alpha^2 p_2(x|\alpha) + \dots + \alpha^M p_M(x|\alpha)}_{\text{Fixed-order calculation}} + \underbrace{\mathcal{O}(\alpha^{M+1})}_{\text{Unknown}}$$

Example: Thrust [Singular parts Becher, Schwartz; [0803.0342](#)]

$$D_A(\tau) = \frac{1}{\tau} \{C_F [-4 \ln \tau - 3]\}$$

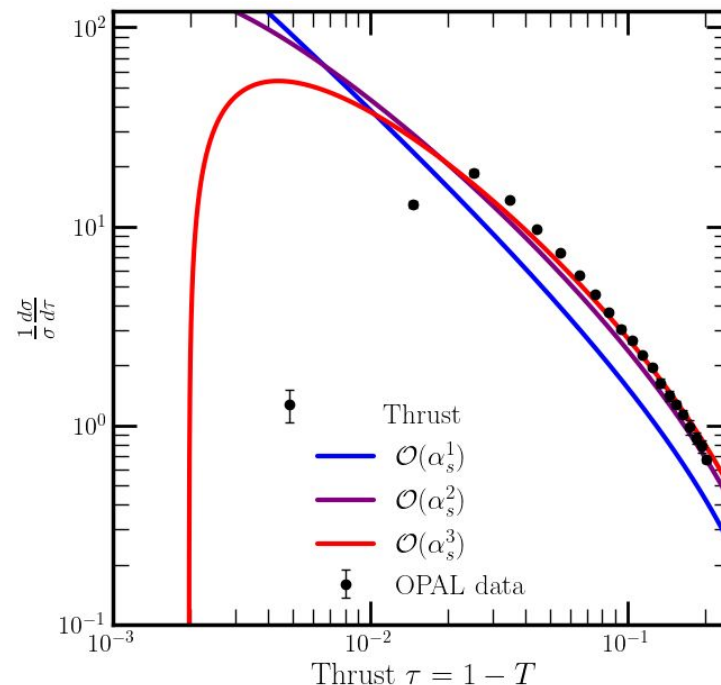
$D_B(\tau)$ This expansion has some pathologies “real life” (all-orders) doesn't have ...

$D_C(\tau)$

1. Non-finite
2. Non-smooth
3. Non-positive
4. Non-normalizable

Large logs can spoil convergence! Lets fix it.

For comparison: **OPAL** Data



Our Solution: The Renormalizing Flow*

Given a Fixed Order prediction for an observable x depending on α

$$p(x|\alpha) = p_0(x) + \alpha p_1(x|\alpha) + \alpha^2 p_2(x|\alpha) + \dots + \alpha^M p_M(x|\alpha) + \mathcal{O}(\alpha^{M+1})$$

Fixed-order calculation – you give us this.

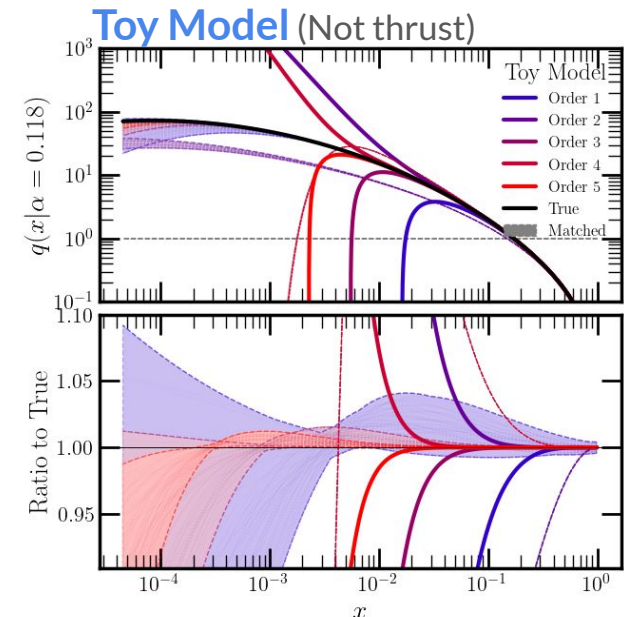
Unknown

We can **guess** the higher order coefficients by using the **Renormalizing Flow Ansatz**, which is parameterized by functions $g_m(t)$:

$$q(t|\alpha) = (g_* \alpha^{m^*}) e^{-\sum_m g_m(t) \alpha^m - \int_0^t dt' (g_* \alpha^{m^*}) \exp\{\sum_m g_m(t') \alpha^m\}}$$

Which is *guaranteed* to fix all of the pathologies:

1. Non-finite → **Finite!**
2. Non-smooth → **Smooth!**
3. Non-positive → **Positive!**
4. Non-normalizable → **Normalized!**



The Renormalizing Flow: Details

1. **Probability Fact:** A positive, smooth, normalized, and finite PDF of a single variable t , where t ranges from 0 to ∞ , can *always* be written as:

$$q(t|\alpha) = f(t, \alpha) e^{-\int_0^t dt' f(t', \alpha)}$$

for *any* function f that is always non-negative and whose integral in t diverges.

2. **Analysis Fact:** All positive functions f that are analytic in α can be written as the *exponential* of a function g , given by*:

$$g(t, \alpha) = -\log\left(g_*(t)\alpha^{m^*}\right) + \sum_{m=0} g_m(t)\alpha^m$$

3. **Scaling:** An observable g transformed into t :

$$t = \log\left(\frac{1}{x}\right) \quad \text{This is where Logs come from!}$$

These three ingredients together lead to our Ansatz!

Matching

$$q(t|\alpha) = (g_* \alpha^{m_*}) e^{-\sum_m g_m(t) \alpha^m - \int_0^t dt' (g_* \alpha^{m_*}) \exp\{\sum_m g_m(t') \alpha^m\}}$$

The **Renormalizing Flow** Ansatz is parameterized by functions $g_m(t)$

For a fixed order calculation up to order M :

For $m \leq M$, perturbative matching requires:

$$g(t, \alpha) = -\log(p_{m_*}(t) \alpha^{m_*}) + \sum_k \frac{1}{k} \left(\left[-\sum_{m>m_*}^M \frac{p_m(t)}{p_{m_*}(t)} \alpha^{m-m_*} \right]^k - \left[\sum_{m=m_*}^M \int_0^t dt' p_m(t') \alpha^m \right]^k \right) + \mathcal{O}(\alpha^{M+1})$$


 We divide out the lowest power of α – this power is m_*

This looks scary ... but this is trivial to implement with **autodiff** ~ **ML magic**!

For $m > M$, choose *anything you want*^{*} for $g_m(t)$: neural networks, random guessing, zero, ...

The choice of higher order $g_m(t)$ is perturbative uncertainty – *all* possible positive, smooth, finite, and normalized completions of the fixed order theory fit here!

We'll treat $g_m(t)$ as polynomials in t for simplicity – *logarithmic expansion in x* .

^{*}One condition – $g_m(t)$ must be negative for large t

Toy Example: Exponential

Consider the “platonic” distribution

$$p_{\text{Ideal}}(t|\alpha) = \alpha \exp(-\alpha t)$$

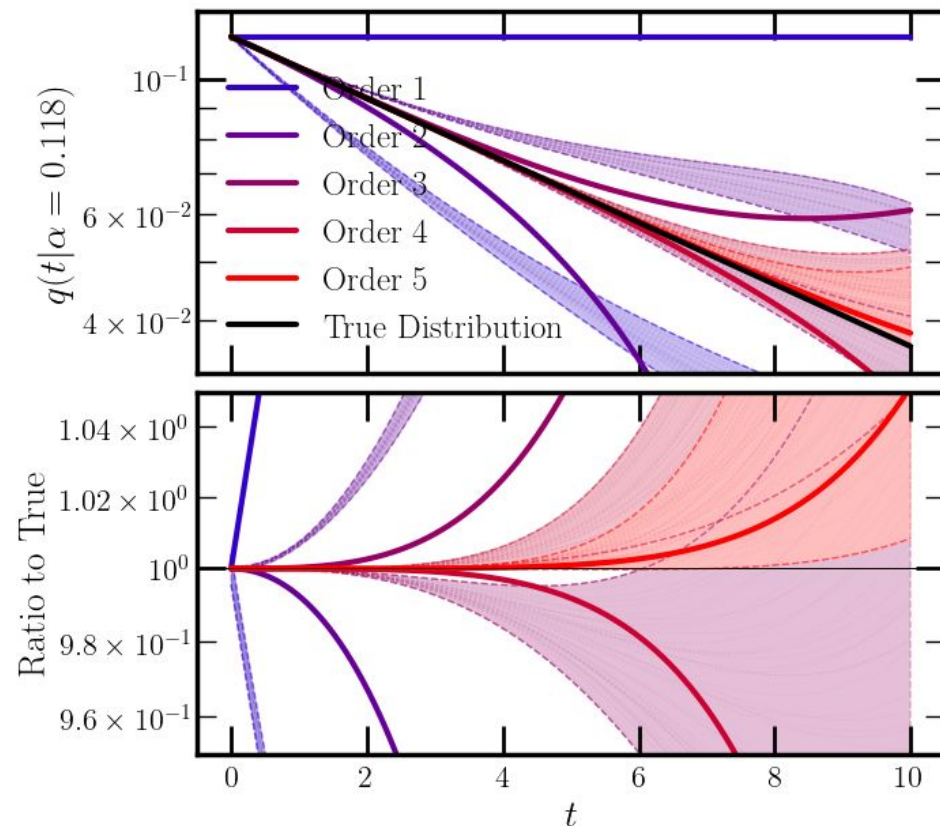
But we only have access to a finite fixed-order expansion

$$p_{\text{FO}}^{(M)}(t|\alpha) = \alpha \sum_m^{M-1} \frac{(-\alpha t)^m}{m!}$$

Let's apply the renormalizing flow!
We'll pick some prior for the higher order coefficients g_{mn} .

Solid: Fixed Order (Pathological)

Bands: Distribution of the Renormalizing Flow for different random choices of g_{mn}



Every curve in the bands is a *non-pathological* valid distribution that matches the FO!

Toy Example: Exponential

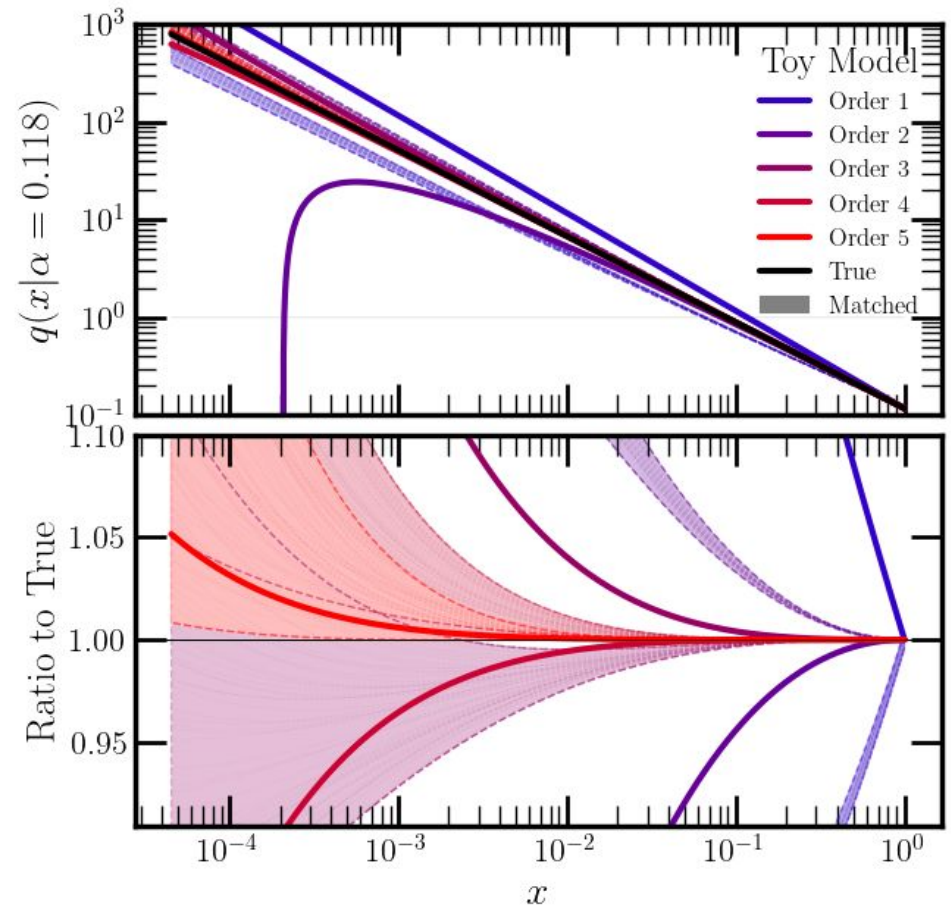
An exponential distribution in t space is a power-law distribution in x space with a slope of α .

The F.O. distributions are only good for large x , and quickly blow up or become negative for small x .

But the Renormalizing Flow *never* will!
It will always be well behaved.

Since the Ansatz is universal, the *true* answer corresponds to *some* choice of higher order g , but we don't know which — but no matter what we guess, our guess is always a valid distribution!

Solid: Fixed Order



Every curve in the bands is a *non-pathological* valid distribution that matches the FO!

Toy Example II: Angularity-Like

Consider the “platonic” distribution

$$p^{\text{LL}'}(\rho) = \frac{\alpha_s C_F}{\pi} \frac{1}{\rho} \left[\log \frac{1}{\rho} \right] \exp \left(-\frac{\alpha_s C_F}{2\pi} \left[\log^2 \frac{1}{\rho} \right] \right)$$

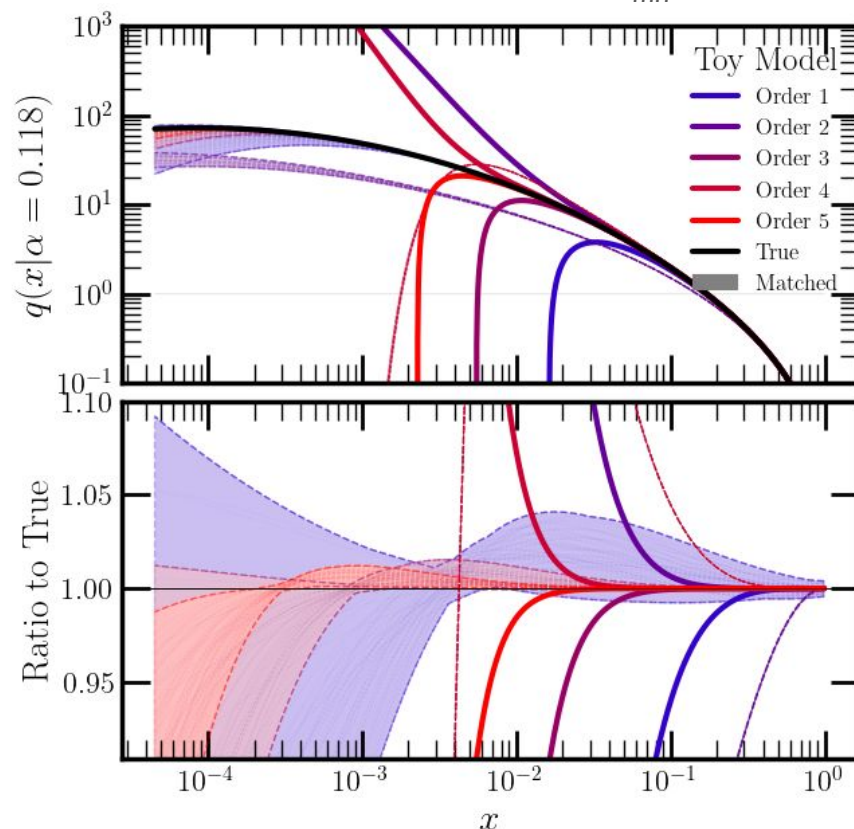
E.g. Let’s pretend that Leading-Log jet angularities are the way the world really works!

We can play the same game, only having access to fixed-order info.

We get *pretty good all-orders* predictions!

Solid: Fixed Order

Bands: Distribution of the Renormalizing Flow for different random choices of g_{mn}



Every curve in the bands is a *non-pathological* valid distribution that matches the FO!

Toy Example III: Angularity-Like + Running

Let's take the same fixed-order predictions from before, but now let α run!

Letting α run effectively just changes higher order coefficients in a complicated way— but *autodiff magic* makes it easy, and we don't have to think about it!

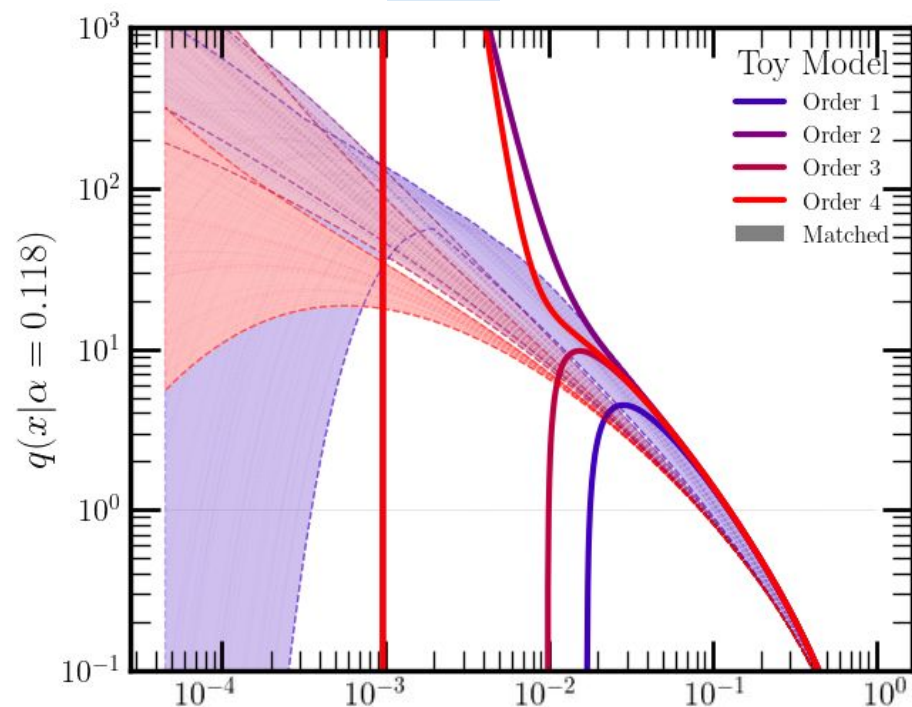
No longer an easy closed-form. We are flying blind!

Weird Fixed Order -> Tamed Renormalizing Flows!

Solid: Fixed Order

Bands: Distribution of the Renormalizing Flow for different random choices of g_{mn}

We still use $\alpha(m_Z)$ to define the conditional distribution $\frac{4\pi}{\alpha_s(\mu_0^2)} - \frac{4\pi}{\alpha_s(Q^2)} = \beta_0 \ln\left(\frac{\mu_0^2}{Q^2}\right)$



Every curve in the bands is a *non-pathological* valid distribution that matches the FO!

CLAIMER

What we are claiming

- **Claiming:** Given a distribution p , pathological or otherwise, we can find a regulated distribution q that is normalized, smooth, positive, and normalized that is perturbatively matched to p .
- **Claiming:** This is a universal approximator: any valid q can be reached with a choice of higher order g , including the “true” one.
- **Claiming:** For particular g 's, we can get meaningful q 's that look like theory calculations and tame large logs!

DISCLAIMER

What we are *not* claiming

- **Not Claiming:** The perturbative structure is *easy* to maintain numerically or that numeric artifacts are manageable.
 - But if you restrict to reasonable choices of g , like $g \sim N(0,1)$, then the results seem reasonable!
- **Not Claiming:** The envelopes give a meaningful uncertainty on higher orders.
 - The envelopes only tell you what *possible* higher order physics is consistent with positivity and normalizability — it *constrains* physics
- **Not Claiming:** This beats logarithmic resummation.
 - This is an all-orders summation. A genuine logarithmic resummation is always better — but for our method, you don't need to know *any* physics!

Key Point: This is *not* a shortcut to doing QCD, just a new way to parameterize things to guarantee nice properties. There is no free lunch, you still have to do physics!

What we are still thinking about...

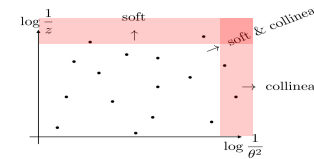
1. Sometimes, we don't have a simple a fixed order expression in closed form. We are currently investigating if it is possible to interface with fixed-order matrix element calculators, potentially in an autodifferentiable way!
2. We have yet to apply this framework to a non-toy physics observable. Hopefully, coming soon!
3. The higher order g 's envelope the true distribution. Can we motivate a prior for them and derive meaningful statistical uncertainties? Can we treat them as nuisance parameters for inference?
4. We only explored 1D distributions in this talk. Is it possible to do more?

Even if you missed some details from each of the four vignettes I went over, I hope you took away:

Part V: Conclusion

Part 0

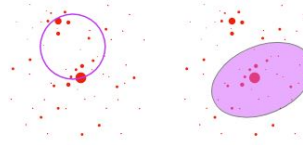
Intro: **IRC-Safety, Jets, and Geometry**



[Ba, Dogra, **RG**, Tasissa, Thaler, [2302.12266](#)]

Part I

Can you Hear the **Shape** of a Jet?



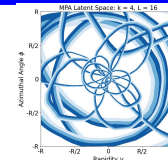
[Ba, Dogra, **RG**, Tasissa, Thaler, [2302.12266](#)]

[**RG**, Thaler, Wu, WIP]

[**RG**, Larkoski, Thaler, [2410.05379](#)]

Part II

Moments of Clarity:
Streamlining Latent Spaces



[**RG**, Osathapan, Thaler, [2403.08854](#)]

Part III

Renormalizing Flows:
Taming Perturbation Theory

$$\begin{aligned} \exp\left[\alpha \frac{d}{d\alpha}\right] p(x) &= p^0(x) + \alpha^1 q^1(x) + \dots + \alpha^N p^N(x) + \mathcal{O}(\alpha^{N+1}) \\ \exp\left[\alpha \frac{d}{d\alpha}\right] q(x) &= q^0(x) + \alpha^1 q^1(x) + \dots + \alpha^N q^N(x) + \mathcal{O}(\alpha^{N+1}) \end{aligned}$$

equal!

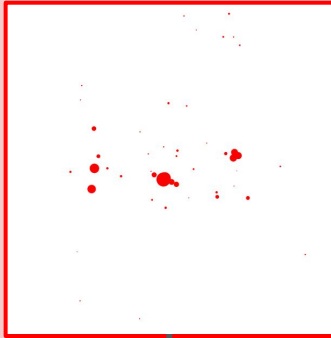
[**RG**, Mastandrea, WIP]

Part V
[You are Here!]

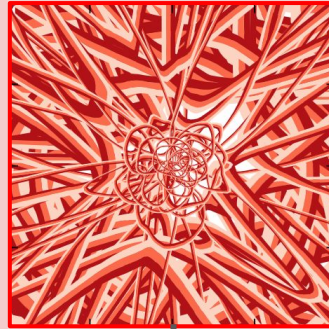
Conclusion

Things about jets a **machine** can understand

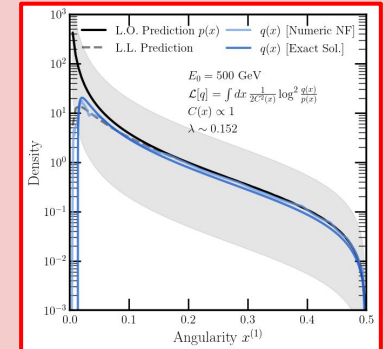
Complicated point clouds



Huge Latent Spaces

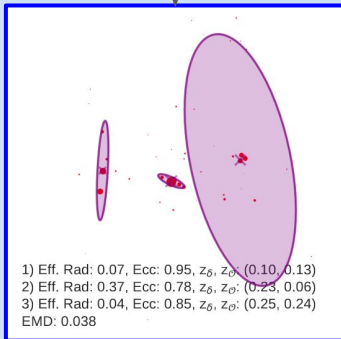


Nontrivially constrained function spaces

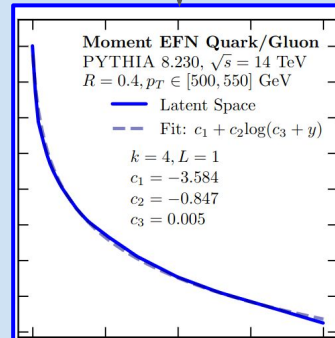


Interface: Targeted and goal-motivated ML function spaces and loss functionals!

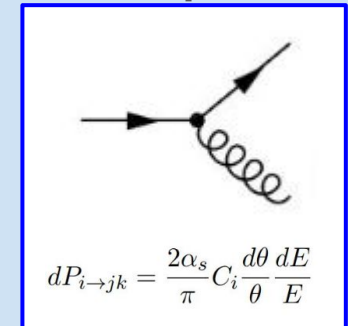
Using faithful
optimal transport



Using Moment
Pooling



Using Re-Normalizing
Flows



Basic Kindergarten Shapes

Addition, multiplication, and a
few elementary functions

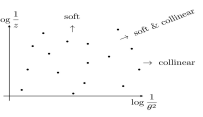
Tree-level Feynman
Diagrams

Things about jets my **mortal physicist brain** can understand

Mail me questions at rikab@mit.edu!

[Ba, Dogra, **RG**, Tasissa, Thaler; [2302.12266](#)]
[**RG**, Thaler, Wu; WIP]
[**RG**, Larkoski, Thaler; [2410.05379](#)]
[**RG**, Osathapan, Thaler; [2403.08854](#)]
[**RG**, Mastandrea; WIP]
[Du Plessis, **RG**; WIP]

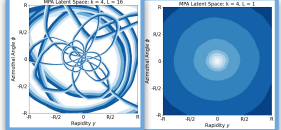
IRC-Safety, Jets, and Geometry



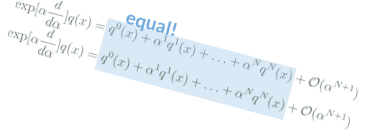
Can you Hear the Shape of a Jet?



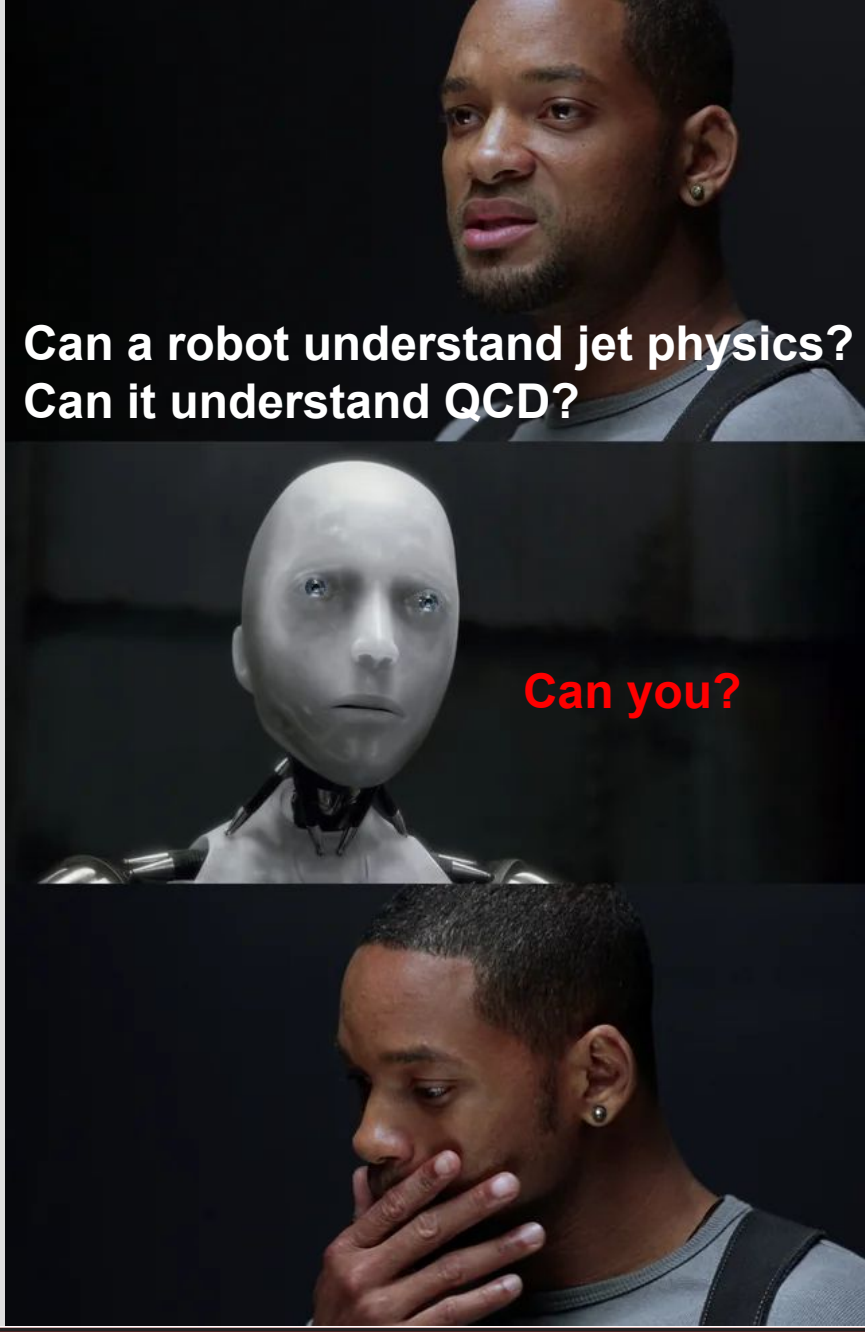
The Moments of Clarity:
Streamlining Latent Spaces



Renormalizing Flows:
Taming Perturbation Theory



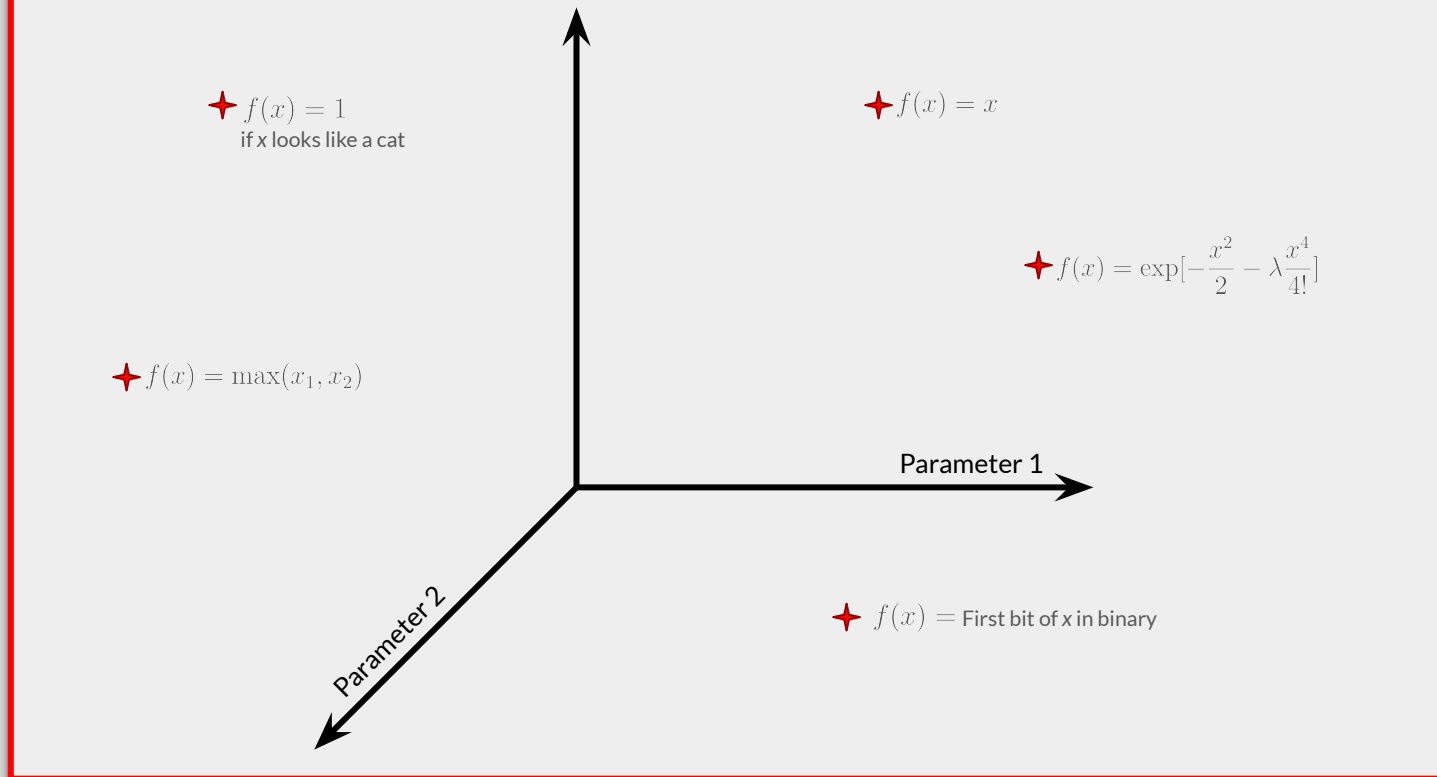
Any Questions?



Backup

What even is Machine Learning (Rikab's Take)?

Ingredient 1: Function Space



Machine Learning (ML) = A method to *numerically* explore the space of functions

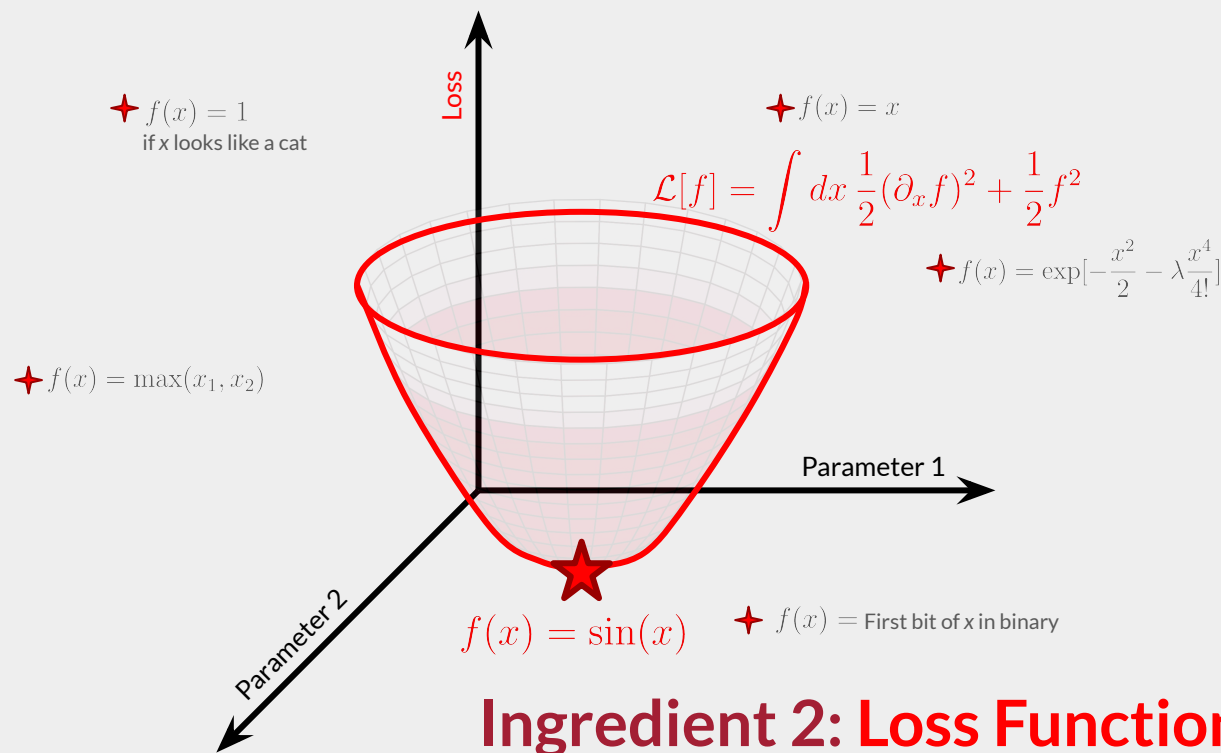
Key workhorse: **Universal Approximation Theorem(s)**

Large classes of functions can be approximated by *Neural Networks* = Parametrized Functions

For this talk, the details of how this approximation is constructed (layers, activations, ...) don't matter!

What even is Machine Learning (Rikab's Take)?

Ingredient 1: Function Space



Ingredient 2: Loss Function(al)

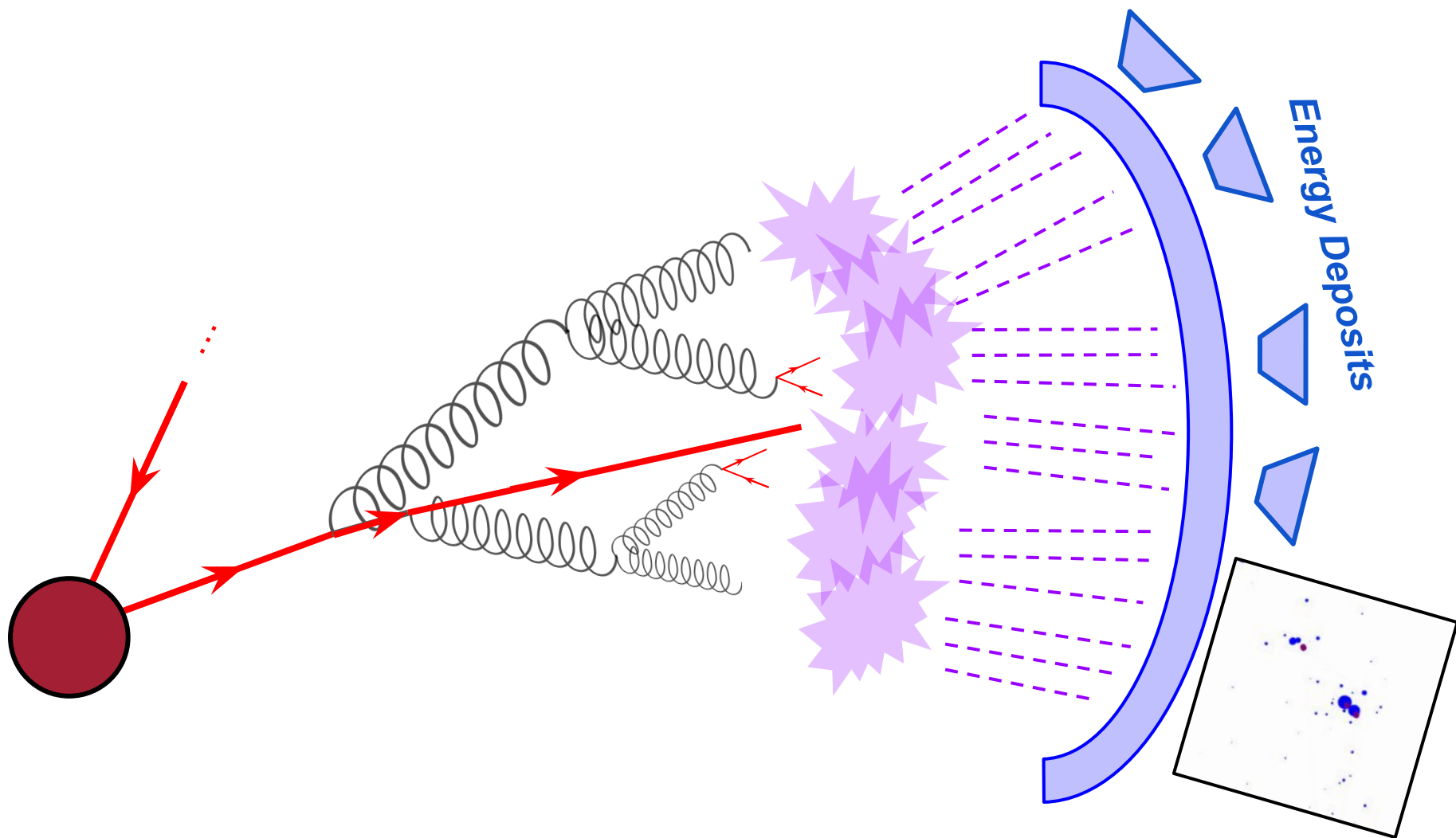
Many problems in life (physics and statistics) involve taking **minimums** over function space!

This is *identical* to how we do Lagrangian Mechanics, down to using the Euler-Lagrange equations.

Statistical Inference (Part I), Defining Observables (Part II), Classification (Part III), even Resummation (Part IV)

ML lets us do functional minimization *numerically*!

The details of how we do this (gradient descent, autodifferentiation, ...) are not important for this talk.



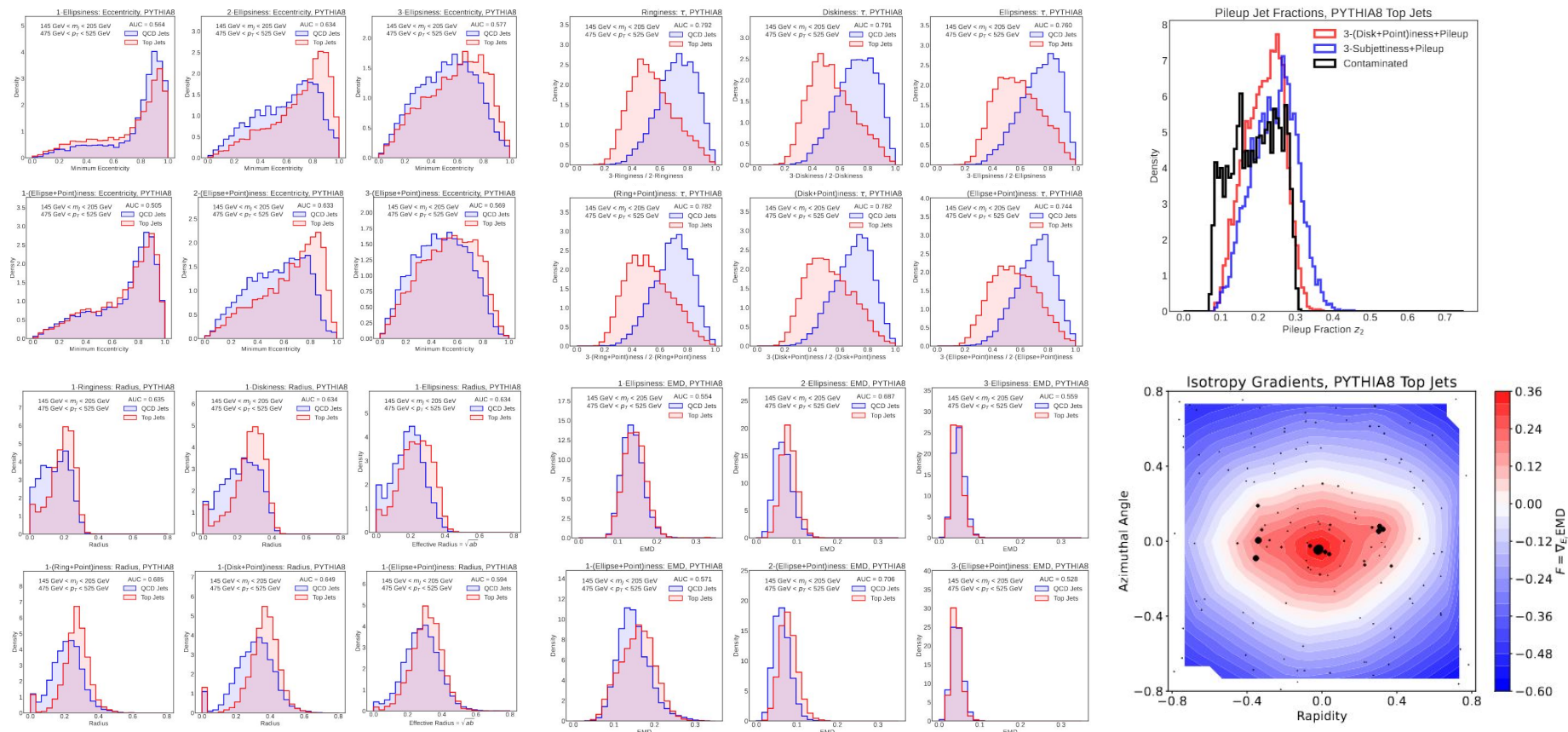
The general strategy for *all* of these problems will be:

1. Identify what properties we want our solution function f to have: *e.g. prior-independent, nice Gaussian limit, contains particular statistical information, IRC-safety, certain geometrical properties, easy function composition, easily plottable limits, physical constraints like smoothness or positivity ...*
2. Design a space of parameterized functions F that manifestly satisfy the properties you want. *e.g. If I want Gaussian functions in z , $\log(T(x,z)) = A(x) + (B(x) - z)C^{-1}(x)(B(x) - z)$ parametrizes all of them!*
3. In conjunction, design a “Loss Functional” $\mathcal{L}[f]$ over F whose minimum satisfies the properties you want. *e.g the Wasserstein Metric guarantees IRC-safety! The DV-loss is prior-free! We know how to do Euler-Lagrange, so just check!*
4. Perform the minimization! *This is where the actual ML comes in – but these are just incidental numeric details!*

If you don't remember the details of each physics example I show in this talk, I want you to remember this overall theme:

Key Takeaway: This is *all you need* for learning physics from machines!

New IRC-Safe Observables



... **Lots** of extractable information!

Automatic Grooming with Shapes

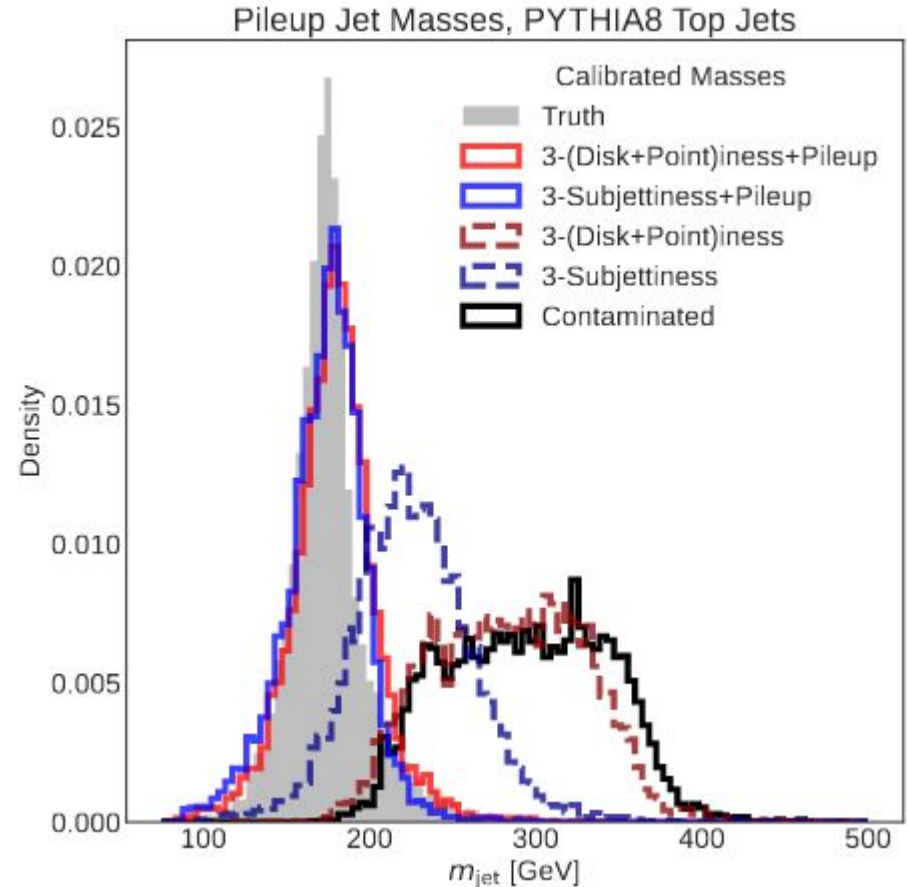
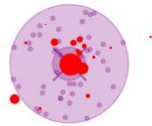
Use shapes to approximate events and extract masses – model pileup with a uniform background with floating weight!

No external hyperparameters, unlike softdrop.
Only need to assume pileup is uniform!

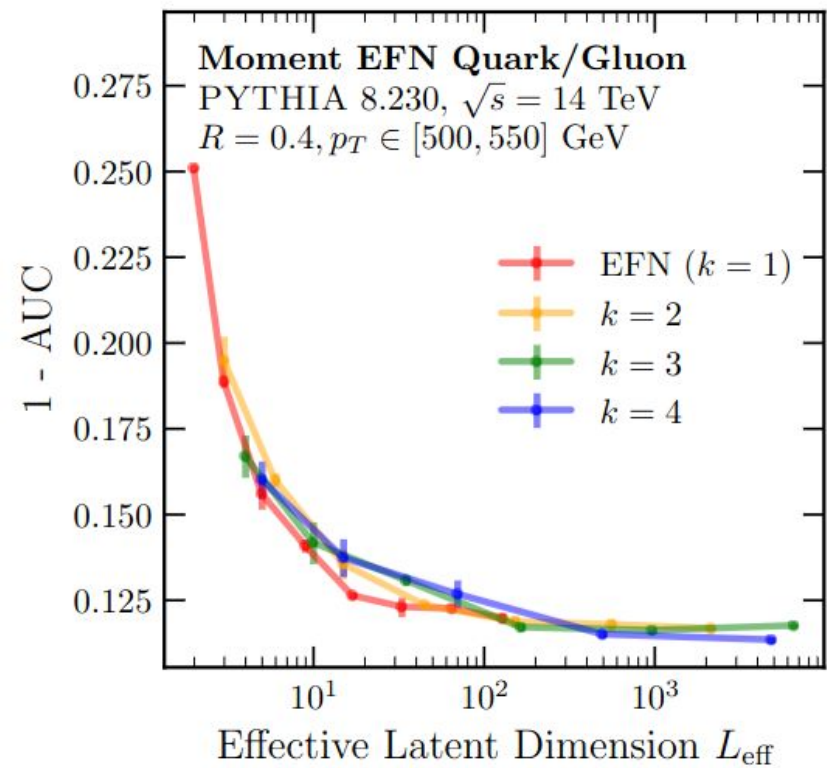
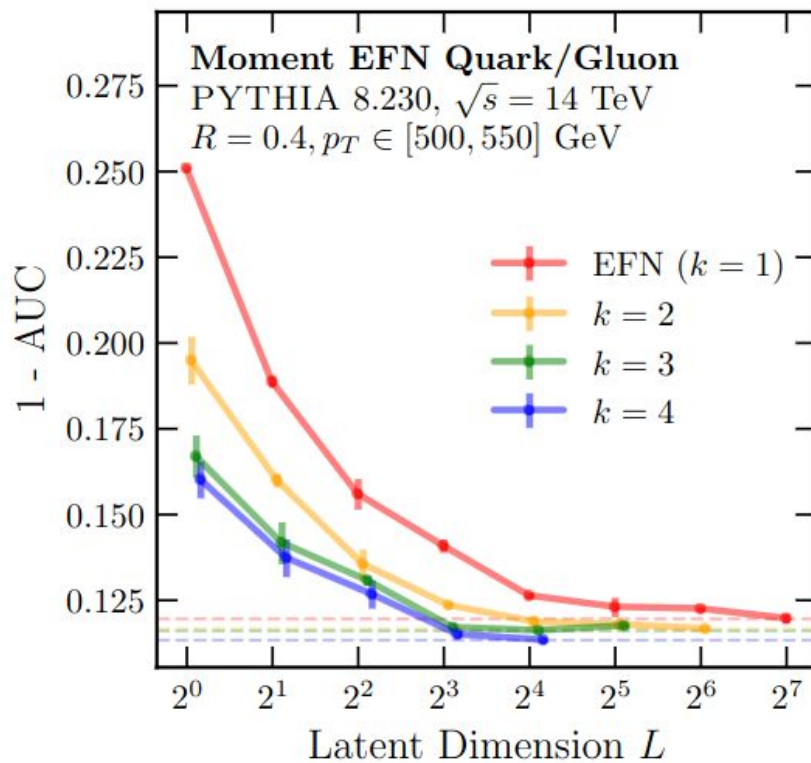
Contaminate top jets with 5-30% extra energy spread uniformly in an 0.8×0.8 plane

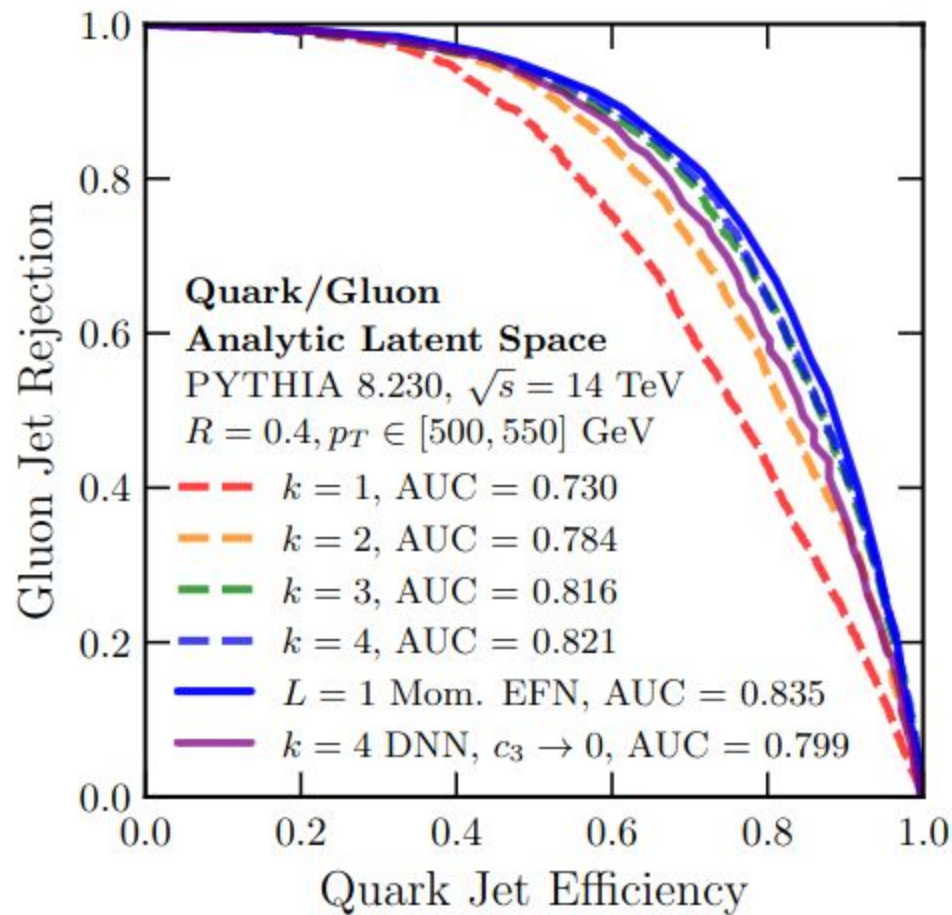
Consider 4 shapes:

- 3-Subjettiness
- 3-Subjettiness + Pileup
- 3-(Disk+Point)iness ←
- 3-(Disk+Point)iness + Pileup

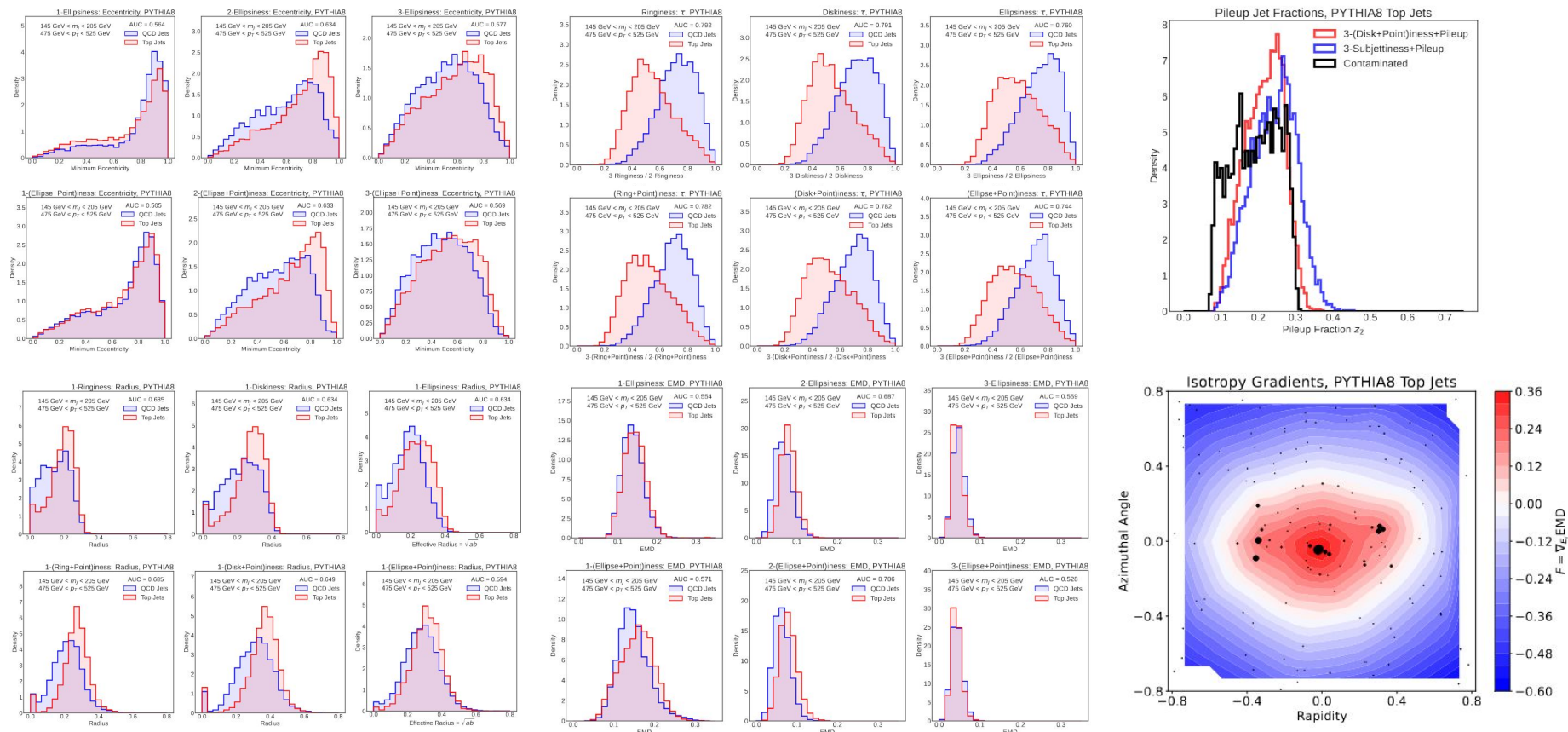


Can also consider ellipses instead of disks – only marginally better performance





New IRC-Safe Observables



... **Lots** of extractable information!

Automatic Grooming with Shapes

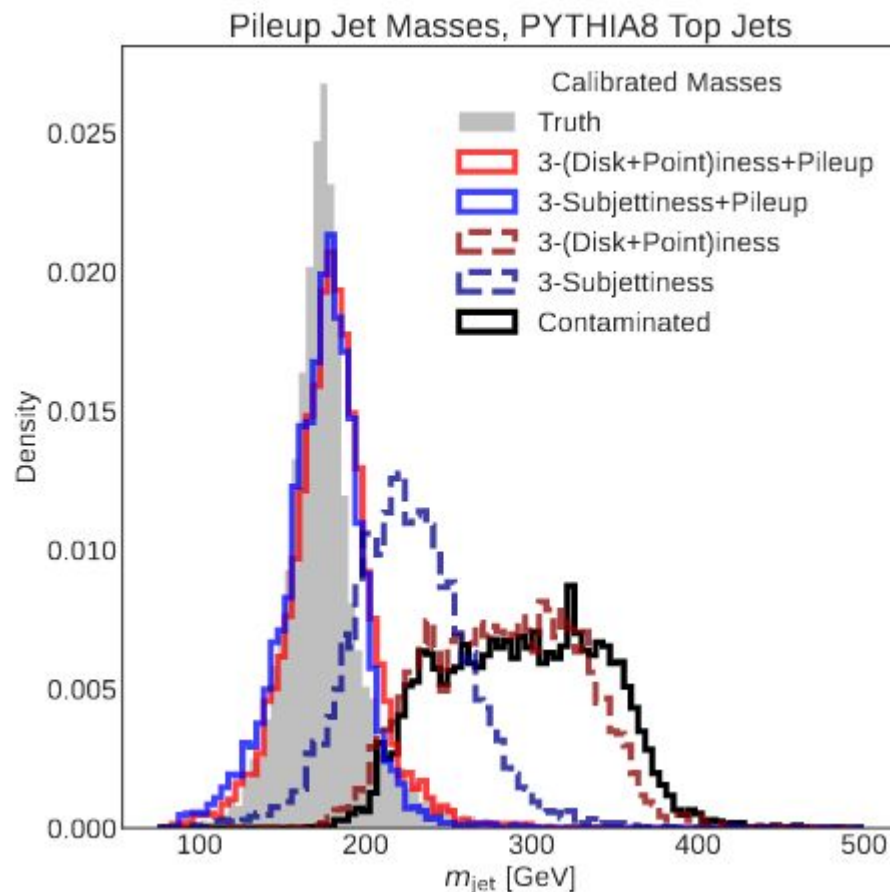
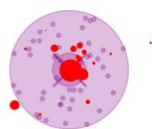
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No external hyperparameters, unlike softdrop.
Only need to assume pileup is uniform!

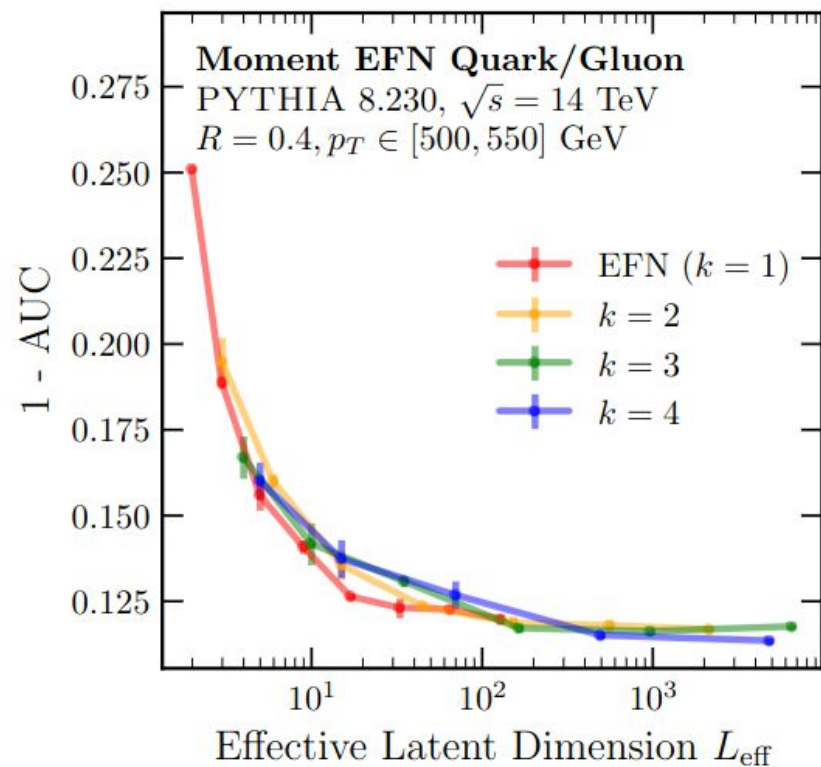
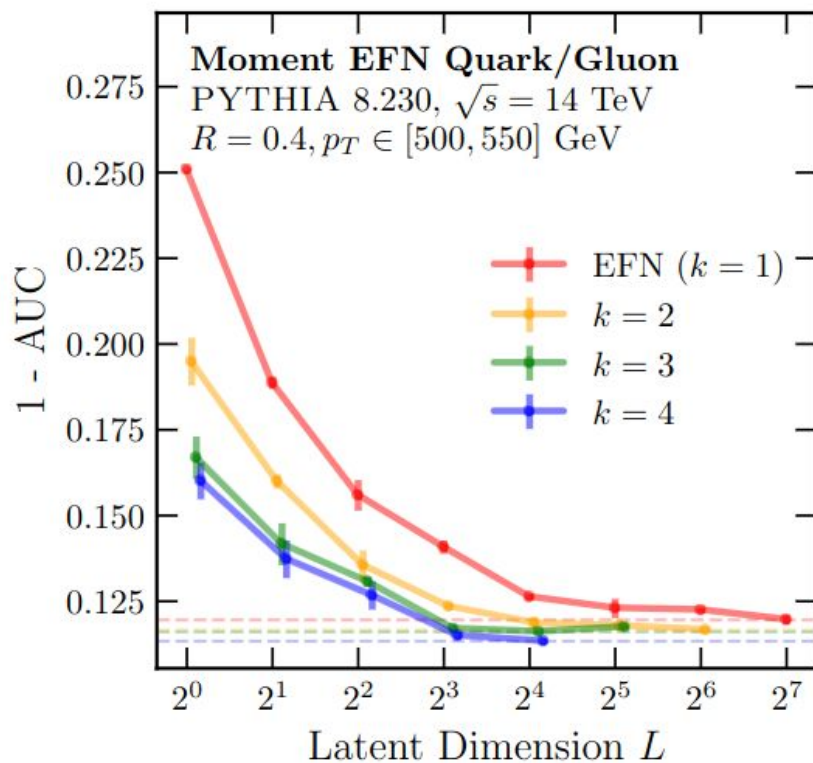
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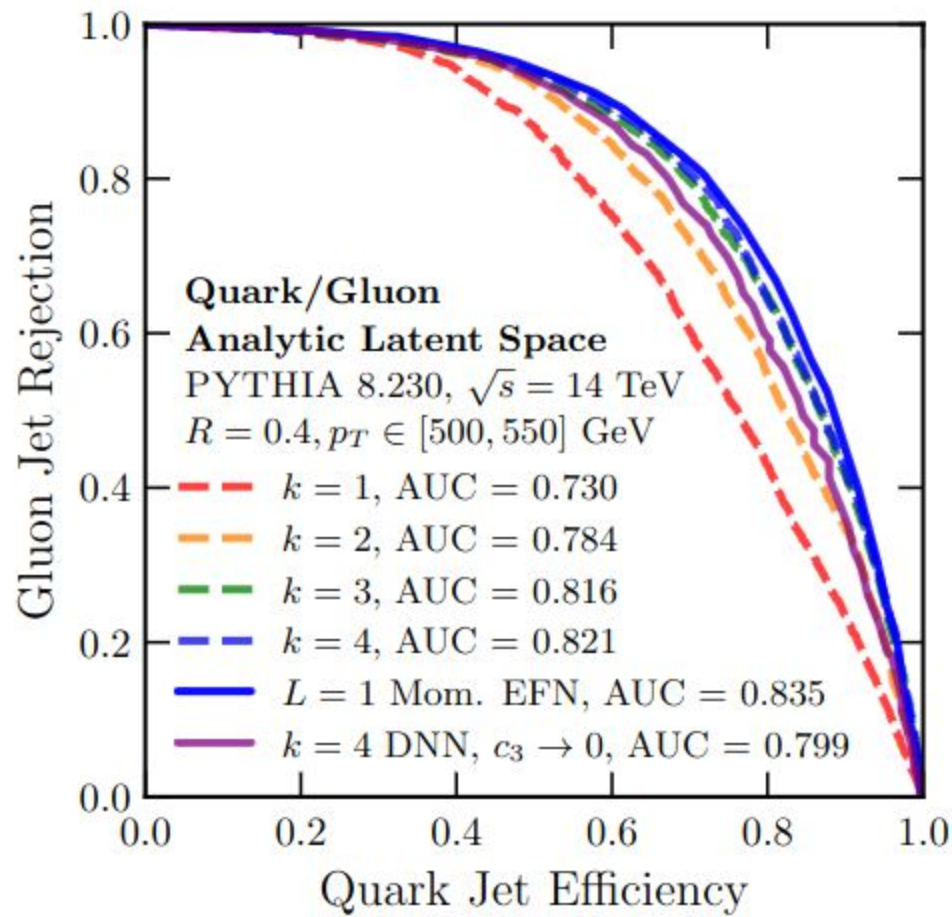
Consider 4 shapes:

- 3-Subjettiness
- 3-Subjettiness + Pileup
- 3-(Disk+Point)iness ←
- 3-(Disk+Point)iness + Pileup



Can also consider ellipses instead of disks – only marginally better performance





The Spectral EMD

Similar to the ordinary EMD, the **Spectral EMD (SEMD)**¹ is an IRC-saf metric between events or jets, computed on their **spectral representations**:

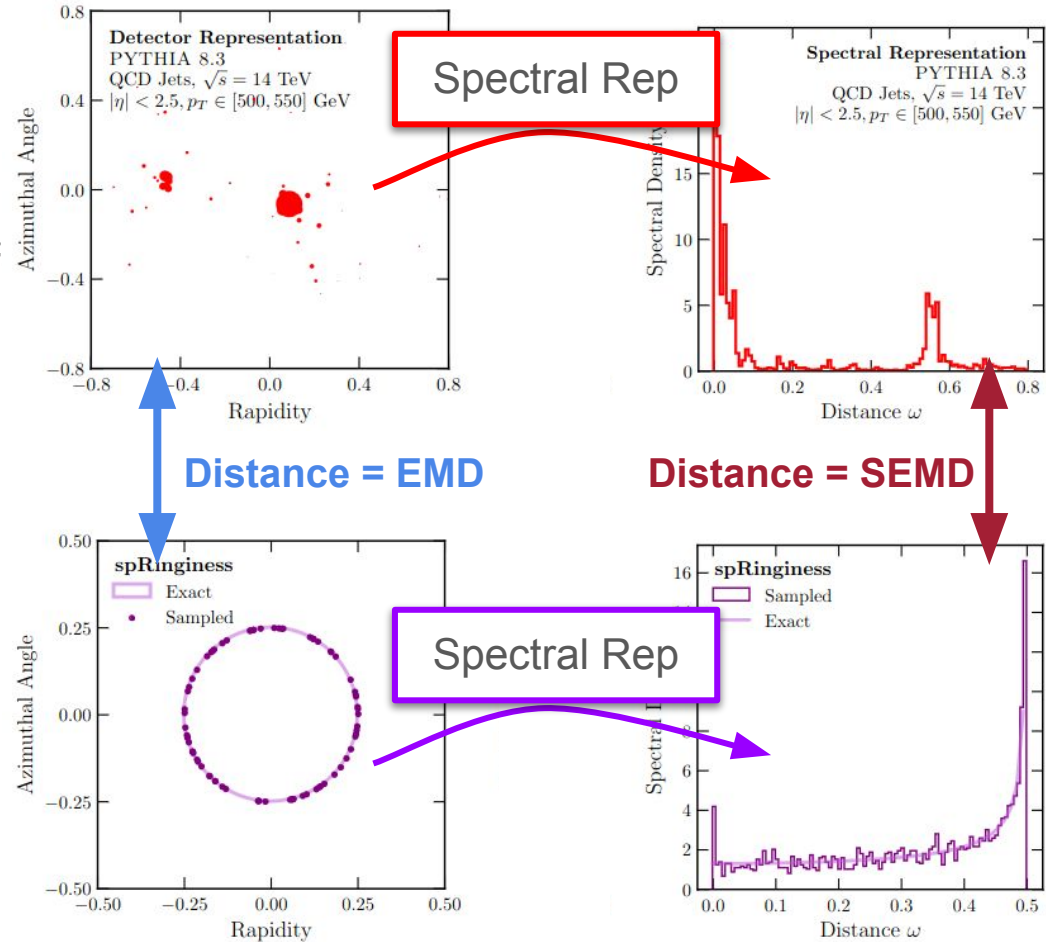
$$s(\omega) \equiv \sum_{i,j \in \mathcal{E}} E_i E_j \delta(\omega - \omega_{ij})$$

= list of energy-weighted pairwise distances

Then the p -SEMD is given by:

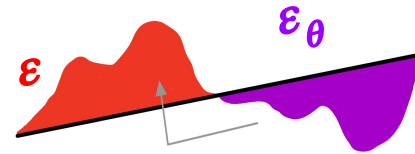
$$\text{SEMD}_p(s_A, s_B) \equiv \int_0^{E_{\text{tot}}^2} dE^2 |S_A^{-1}(E^2) - S_B^{-1}(E^2)|^p$$

Inverse of integral of $s(\omega)$



The SEMD automatically respects all isometries of the metric ω . In this case, the SEMD is invariant under rotations or translations of either event or jet

Technical Details ...



For $p = 2$, possible problem on the sp

For events A, B , the p **spectral EMD** is defined as (1D OT!):

EMD = Work done to move "dirt" optimally

$$\text{SEMD}_{\beta,p}(s_A, s_B) \equiv \int_0^{E_{\text{tot}}^2} dE^2 |S_A^{-1}(E^2) - S_B^{-1}(E^2)|^p$$

$$\begin{aligned} \text{SEMD}_{\beta,p=2}(s_A, s_B) = & \sum_{i < j \in \mathcal{E}_A} 2E_i E_j \omega_{ij}^2 + \sum_{i < j \in \mathcal{E}_B} 2E_i E_j \omega_{ij}^2 \\ & - 2 \sum_{n \in \mathcal{E}_A^2, l \in \mathcal{E}_B^2} \omega_n \omega_l (\min[S_A(\omega_n^+), S_B(\omega_l^+)] - \max[S_A(\omega_n^-), S_B(\omega_l^-)]) \\ & \times \Theta(S_A(\omega_n^+) - S_B(\omega_l^-)) \Theta(S_B(\omega_l^+) - S_A(\omega_n^-)), \end{aligned}$$

S = **cumulative spectral function**

$\pm$ indicates whether or not to include ω in the sum

The trick: Sum over pairs n of particles within each event.

Looks like $O(N^4)$, but with clever sorting & indexing in 1D, reduces to **$O(N^2)$** !

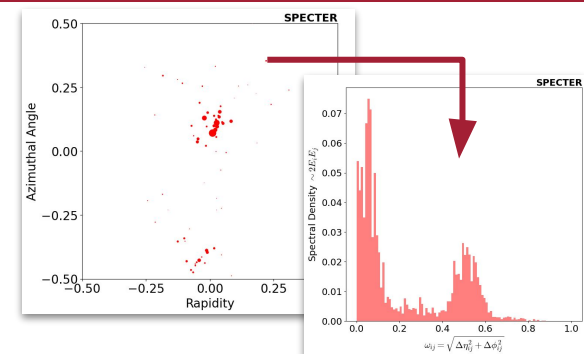
The **spectral density function**

$$s(\omega) = \sum_{i=1}^N \sum_{j=1}^N E_i E_j \delta(\omega - \omega(\hat{n}_i, \hat{n}_j))$$

Pairwise Distances

Reduces events to 1D, while preserving all* information about the event, up to translations and rotations.

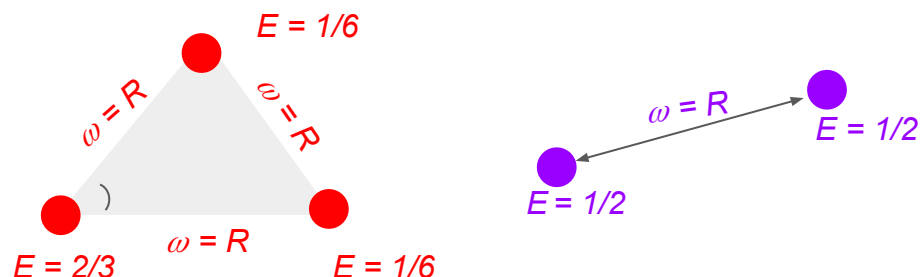
*up to measure 0, but important degeneracies, ask me about this later!



EMDs, ...

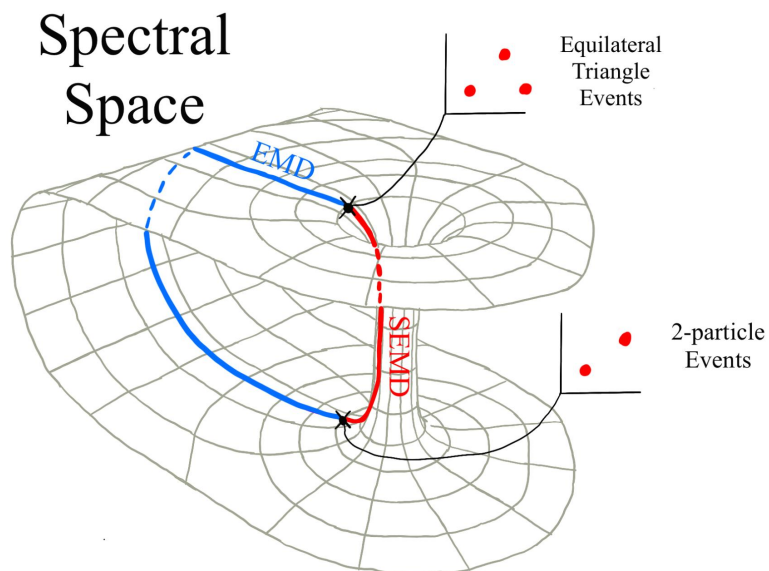
The EMD vs. The SEMD

The SEMD is *topologically different* from the ordinary EMD!



This three particle event looks very different from this two particle event, and thus they have a large EMD.

But their SEMD is zero! The spectral function only cares about pairwise distances and degenerate configurations can occur.

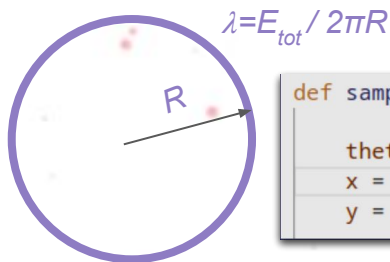


The space of events gets “pinched” at degenerate configurations when looking at only their spectral representations

This can come into play when events have 3 hard prongs, e.g. top jets!

Full Example: How “ring-like” are jets?

Step 1: Define the shape with parameters

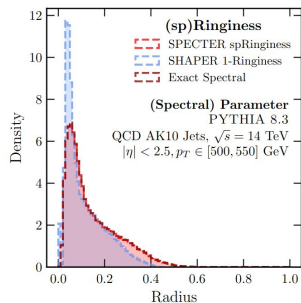
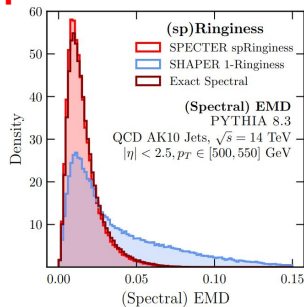


```
def sample_circle(params, N, seed):
    thetas = jax.random.uniform(seed, shape=(N,))
    x = params["Radius"] * jnp.cos(thetas)
    y = params["Radius"] * jnp.sin(thetas)
```

Unlike ordinary EMD, not necessary to specify center / orientation!

Pictured: 100k Jets, PYTHIA 8 QCD Jets

Step 5: Plots!

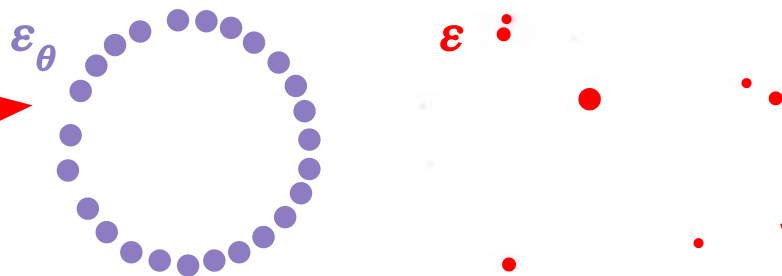


SPECTER

Our code framework
for these calculations



Step 2: Sample from Parameterized Shapes

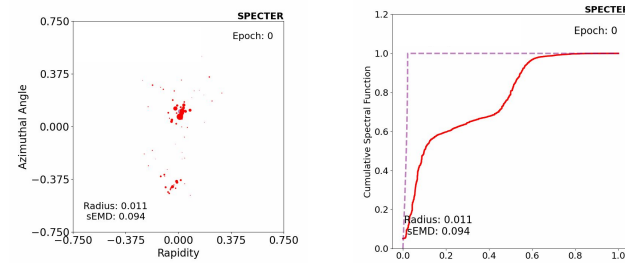


Step 3: Calculate the spectral metric between events and shapes

$$\text{SEMD}_{\beta, p=2}(s_A, s_B) = \sum_{i < j \in \mathcal{E}_A} 2E_i E_j \omega_{ij}^2 + \sum_{i < j \in \mathcal{E}_B} 2E_i E_j \omega_{ij}^2 - 2 \sum_{n \in \mathcal{E}_A^2, l \in \mathcal{E}_B^2} \omega_n \omega_l (\min[S_A(\omega_n^+), S_B(\omega_l^+)] - \max[S_A(\omega_n^-), S_B(\omega_l^-)]) \times \Theta(S_A(\omega_n^+) - S_B(\omega_l^-)) \Theta(S_B(\omega_l^+) - S_A(\omega_n^-))$$

Key difference from previous work: We use the SEMD, *not* the EMD!

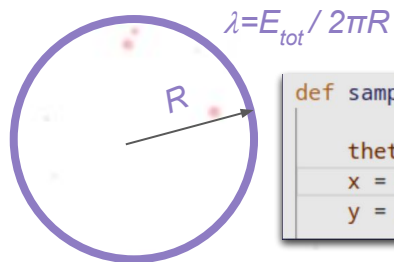
Step 4: Minimize w.r.t. parameters using grads



Pictured: Animation of optimizing for the radius R

Full Example: How “ring-like” are jets?

Step 1: Define the shape with parameters



Unlike ordinary EMD, not necessary to

```
def samp
    theta
    x = p
    y = p
```

The spectral EMD, and its optimization, are often partially or **completely solvable** in closed form!

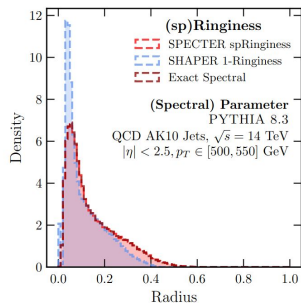
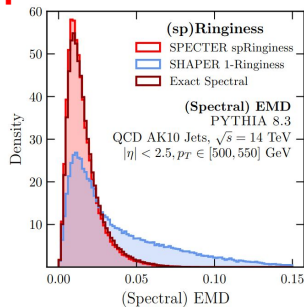
$$R_{\text{opt}} = \frac{2}{\pi} \sum_{\substack{n \in \mathcal{E}^2 \\ \omega_n < \omega_{n+1}}} \omega_n \left[\sin \left(\frac{\pi}{2E_{\text{tot}}^2} \sum_{\substack{n \leq m \in \mathcal{E}^2 \\ \omega_m < \omega_{m+1}}} (2EE)_m \right) - \sin \left(\frac{\pi}{2E_{\text{tot}}^2} \sum_{\substack{n < m \in \mathcal{E}^2 \\ \omega_m < \omega_{m+1}}} (2EE)_m \right) \right]$$

$$\text{SEMD}_{\beta, p=2}(s, s_{\text{jet ring}}) = \sum_{i < j \in \mathcal{E}} 2E_i E_j \omega_{ij}^2 - 2E_{\text{tot}}^2 R_{\text{opt}}^2$$

For many shapes, we can completely short circuit having to perform expensive optimization over an optimal transport problem entirely!

Pictured: 100k Jets, PYTHIA 8 QCD Jets

Step 5: Plots!



SPECTER

Our code framework for these calculations



Step 2: Sample from Parameterized Shapes

Alternatively...

the spectral metric between

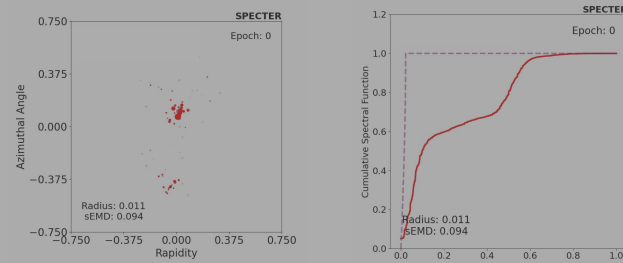
$$2E_i E_j \omega_{ij}^2 + \sum_{i < j \in \mathcal{E}_B} 2E_i E_j \omega_{ij}^2$$

$$(\omega_n^+, S_B(\omega_l^+)) - \max [S_A(\omega_n^-), S_B(\omega_l^-)]$$

$$- S_B(\omega_l^-)) \Theta (S_B(\omega_l^+) - S_A(\omega_n^-)) ,$$

Key difference from previous work: We use the SEMD, *not* the EMD!

Step 4: Minimize w.r.t. parameters using grads



Pictured: Animation of optimizing for the radius R

Hearing Jets (sp)Ring

Runtimes (NVIDIA A100 GPU):

SHAPER (EMD): ~ 36 hours / 100k events

Generalized SPECTER: ~55 seconds / 100k events

Closed Form SPECTER: ~ < 0.1 seconds / 100k events

The SEMD and EMD are qualitatively different, but give similar radii!

They probe the same event length scale

